

LAB2

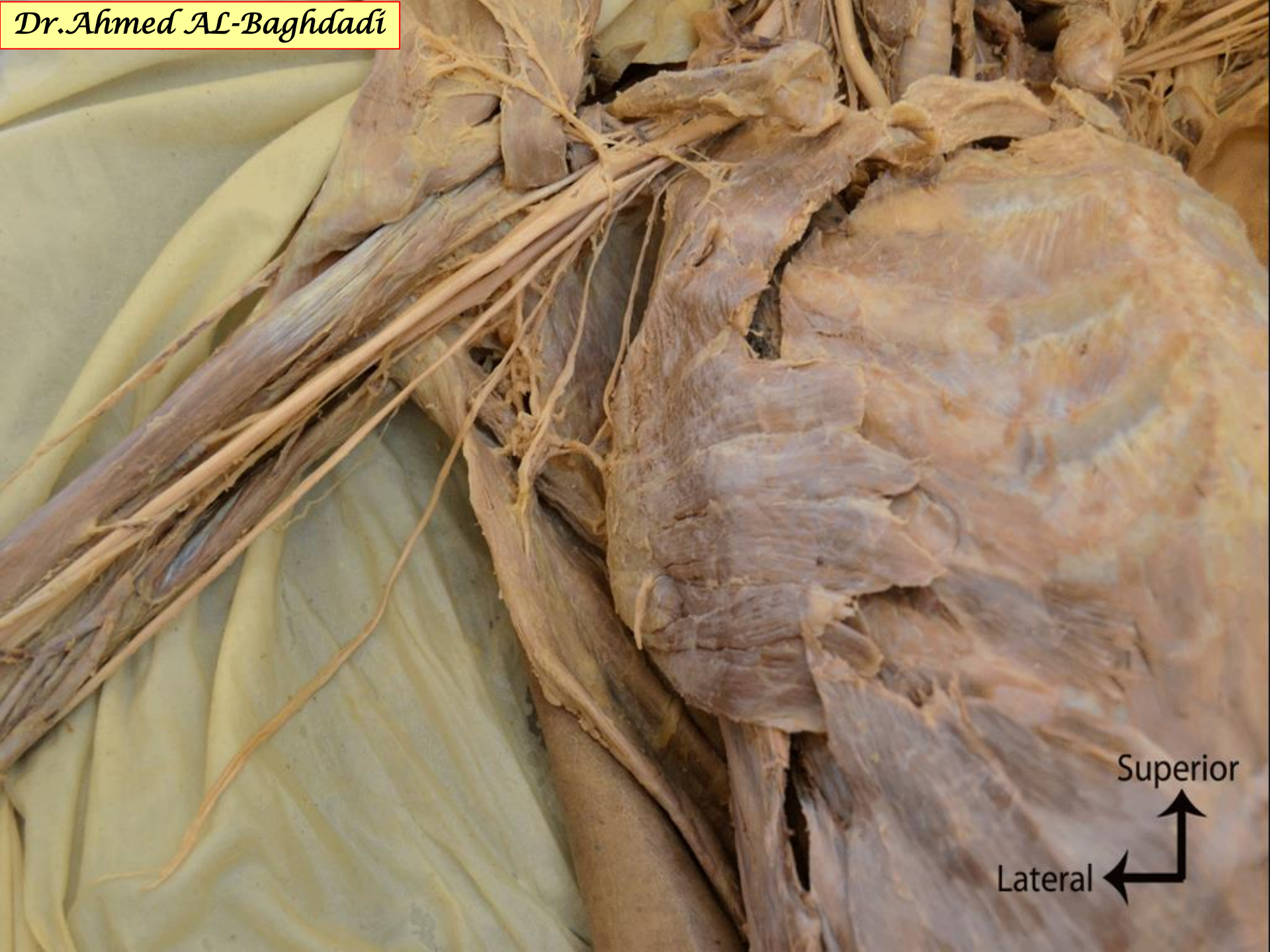
ARM

Dr.Ahmed AL-Baghdadi





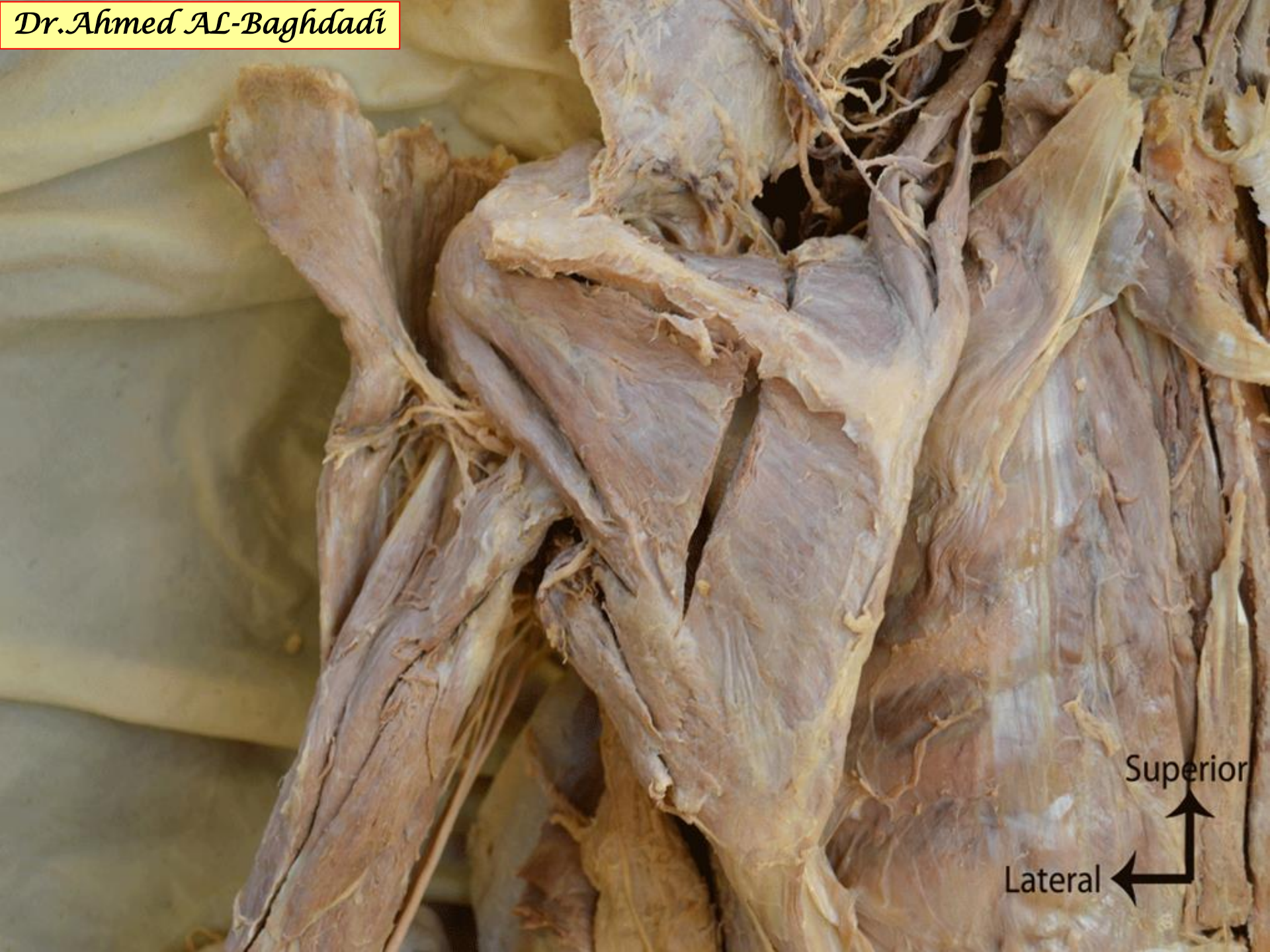




Superior

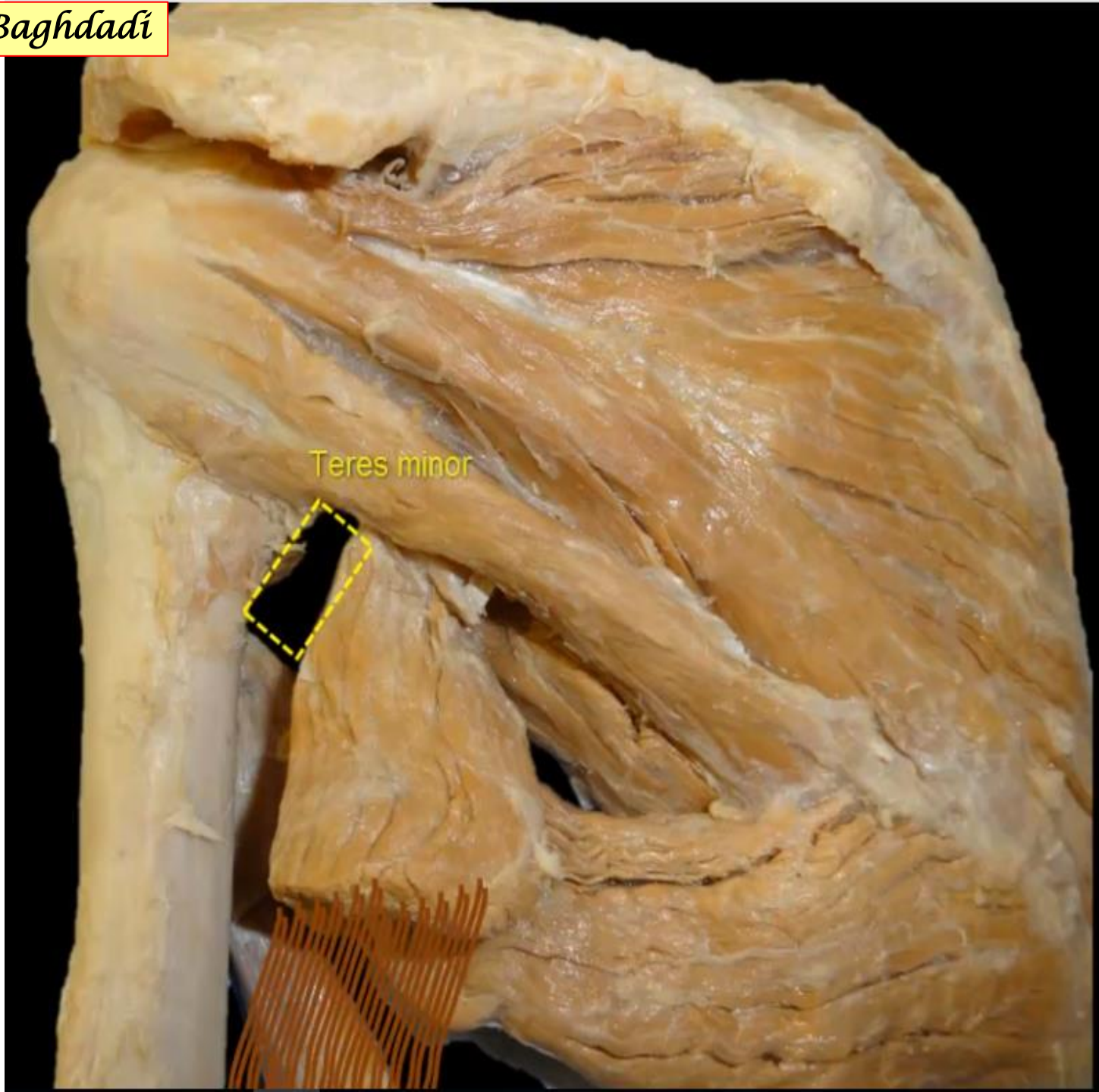
Lateral

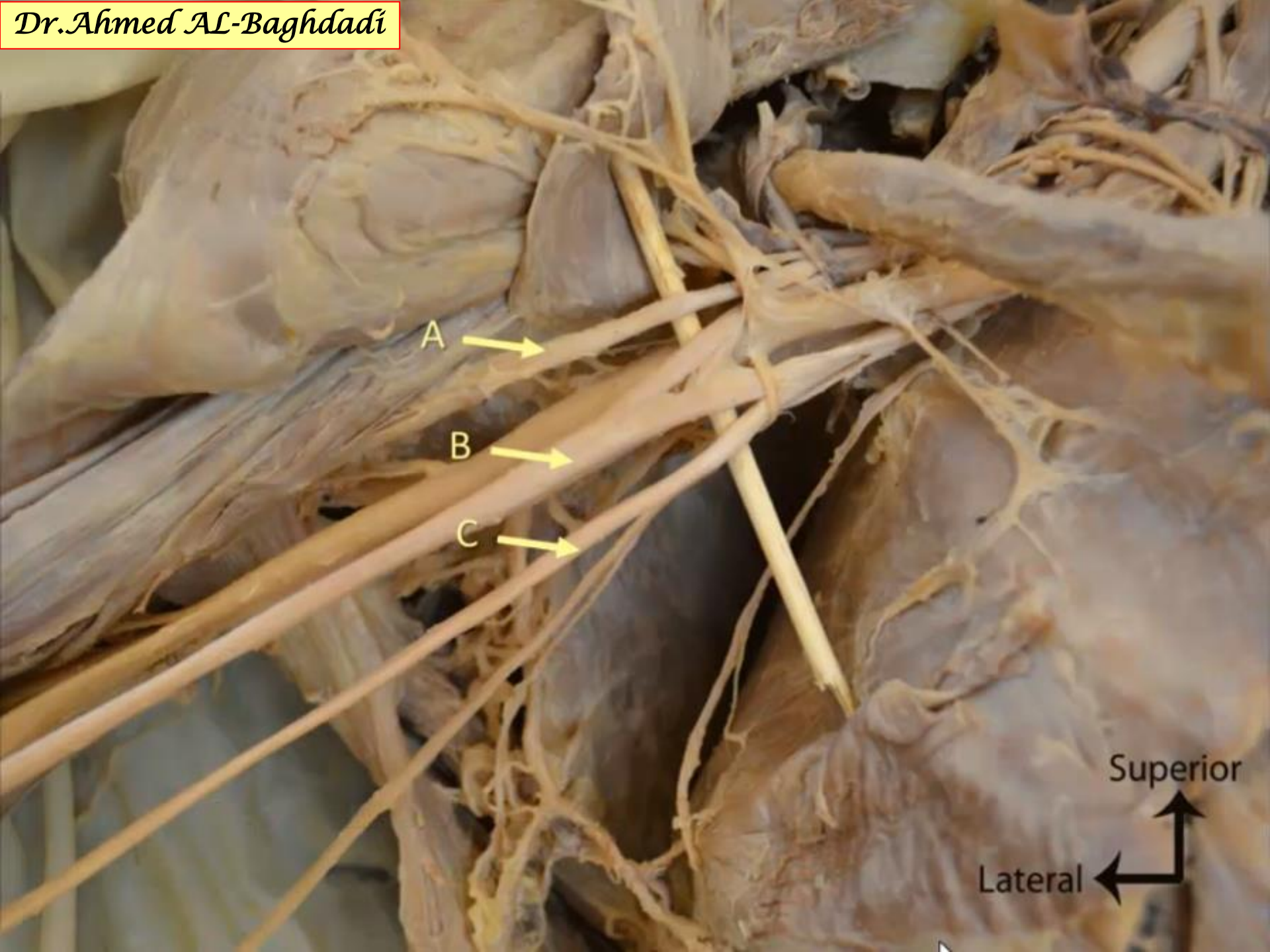




Superior

Lateral







Superior
Lateral

P.Mi

P.M

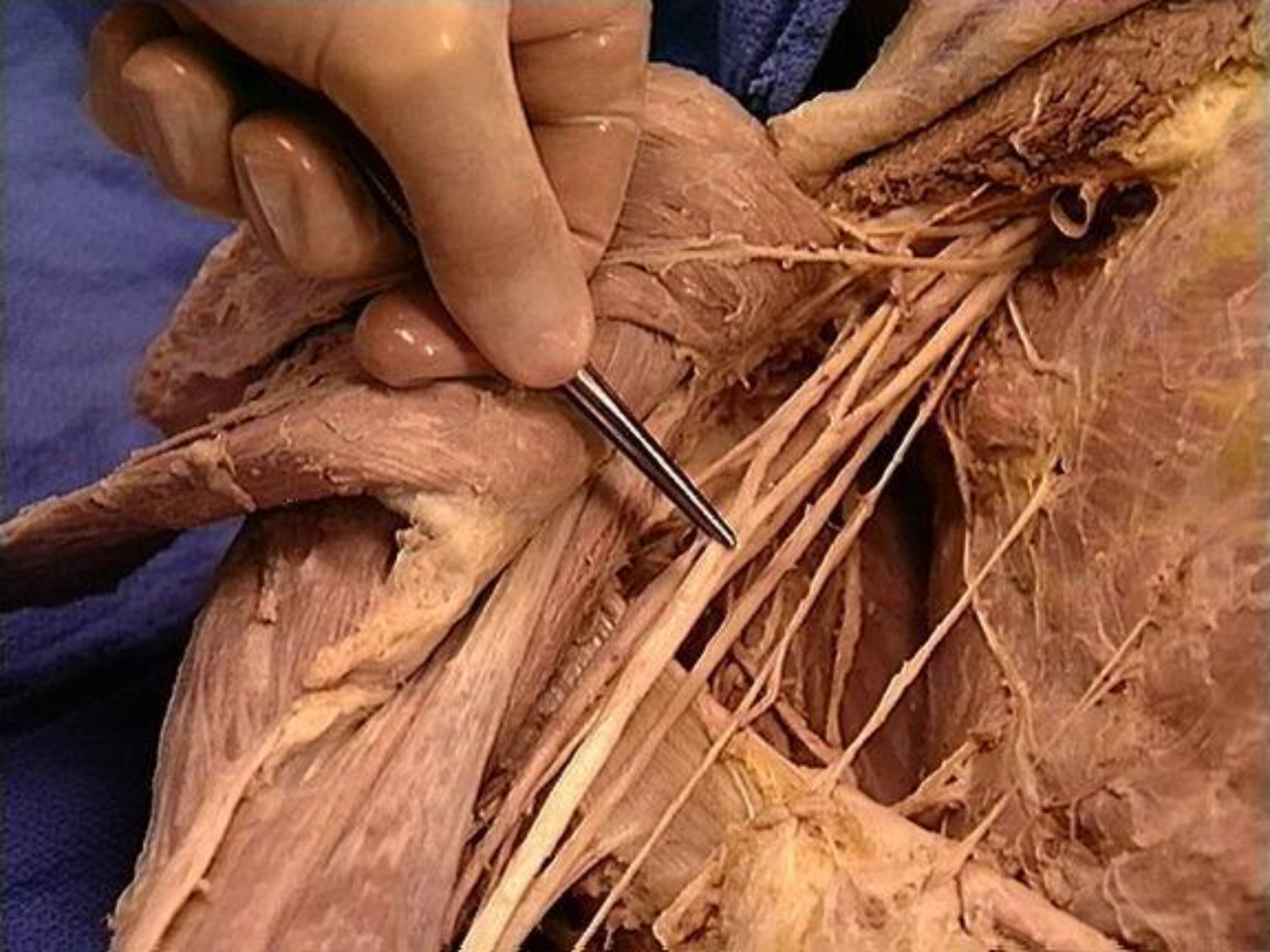
Serratus Ant.

Lat. dorsi

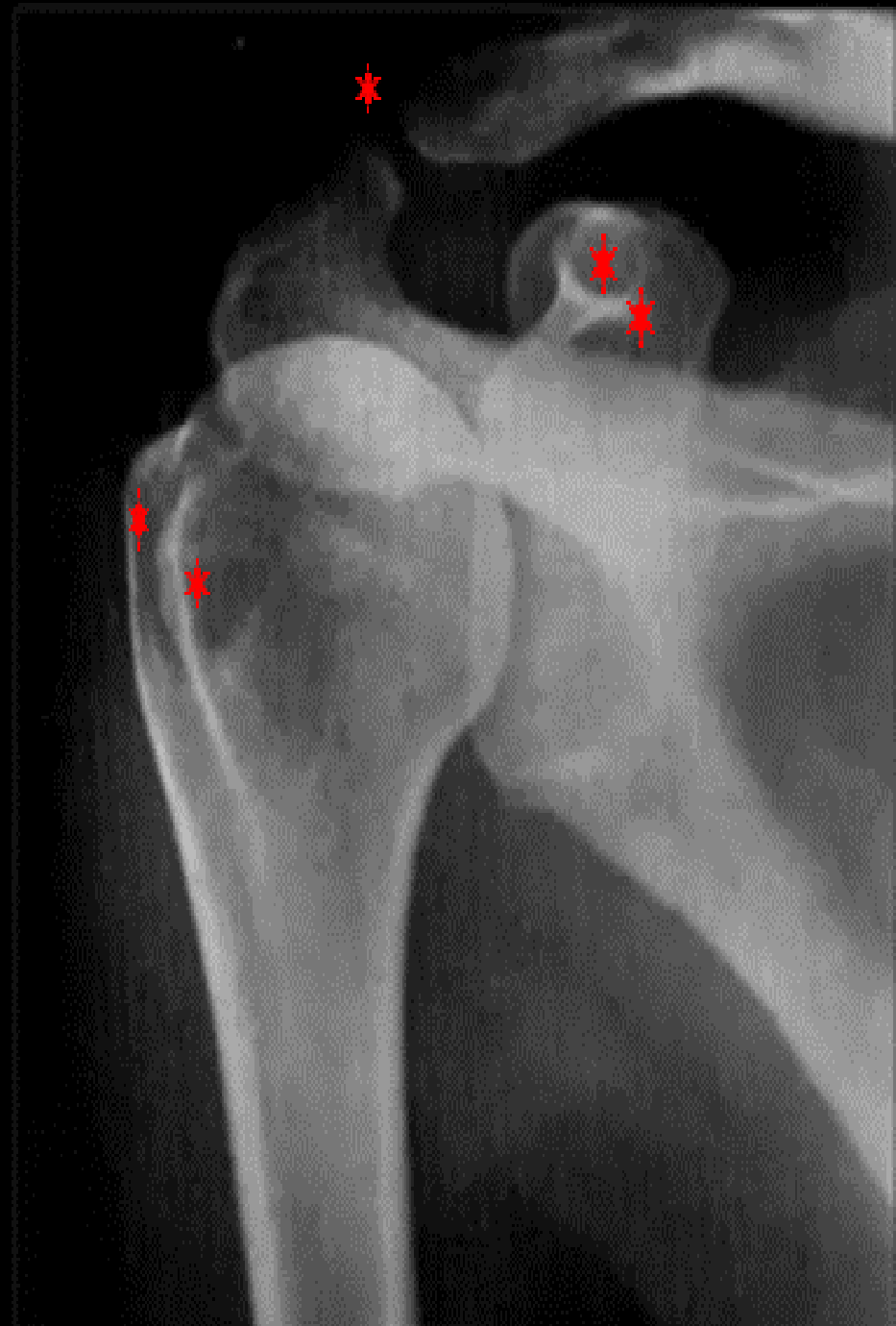
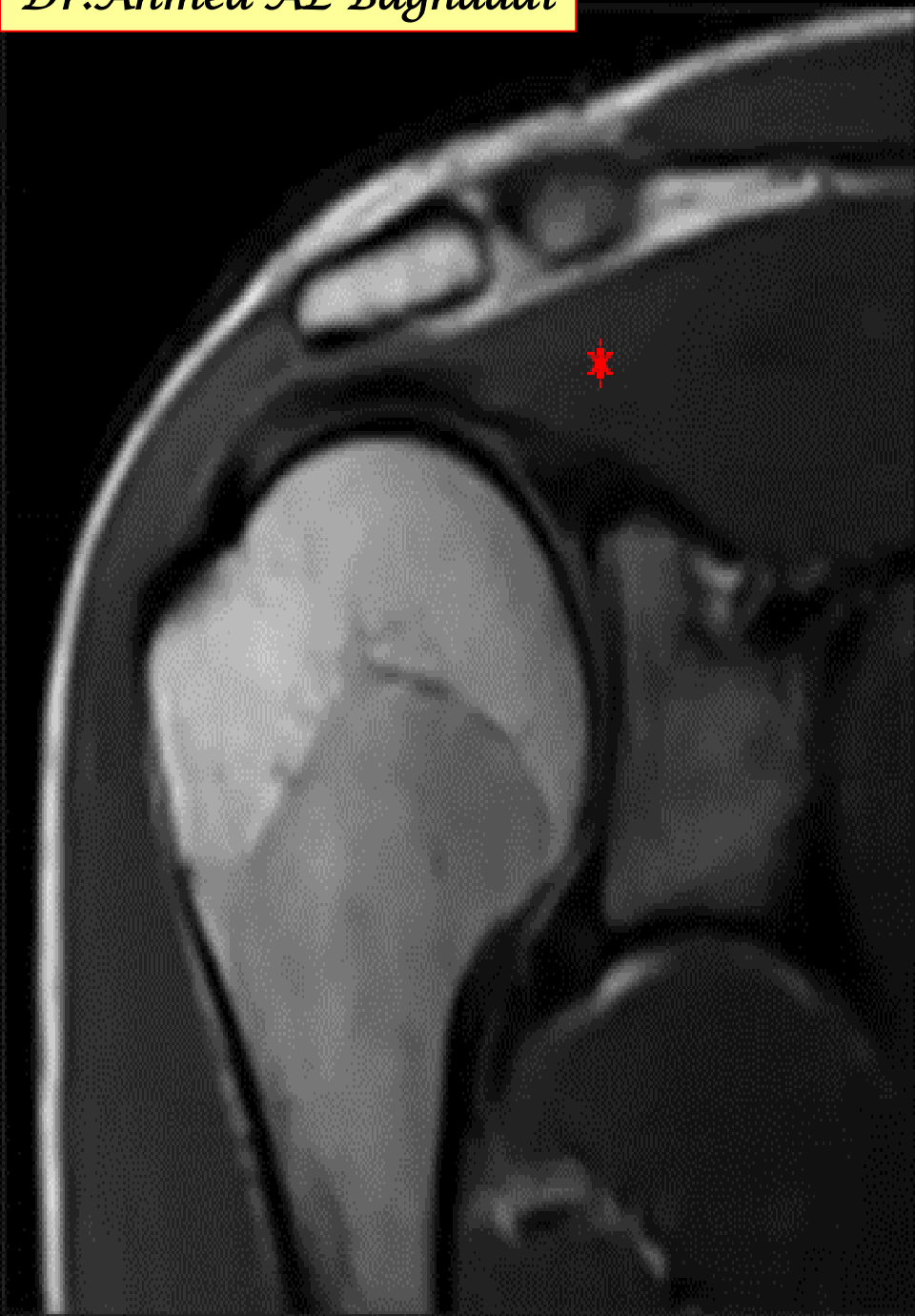
*Short head of Biceps
& Coracobrachialis*





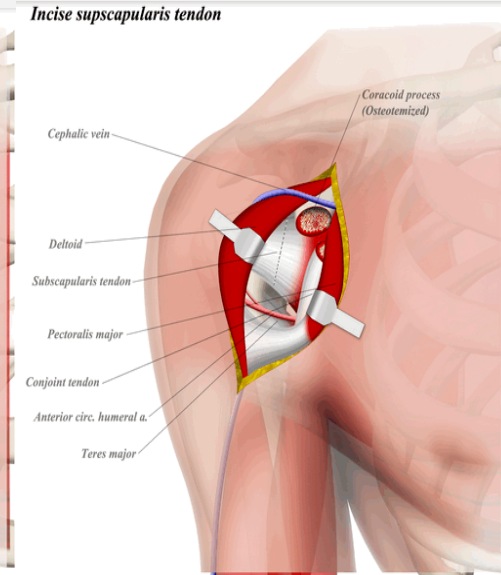
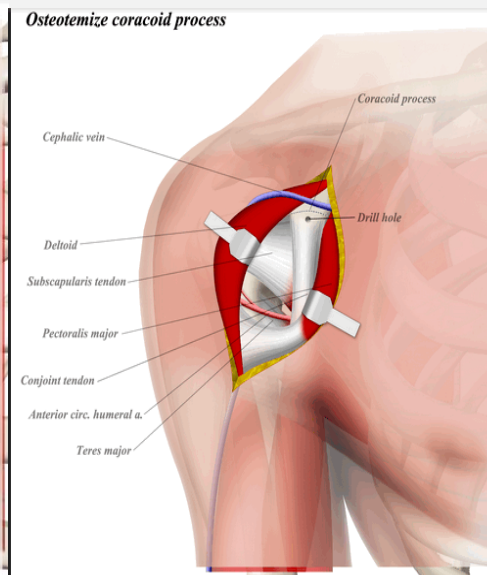
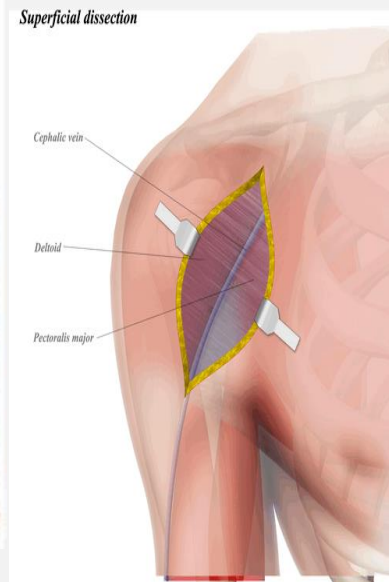
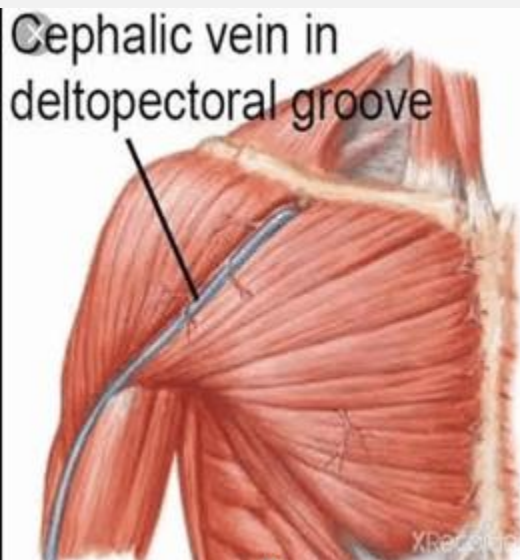






□ describe the deltopectoral approach to the shoulder joint ?

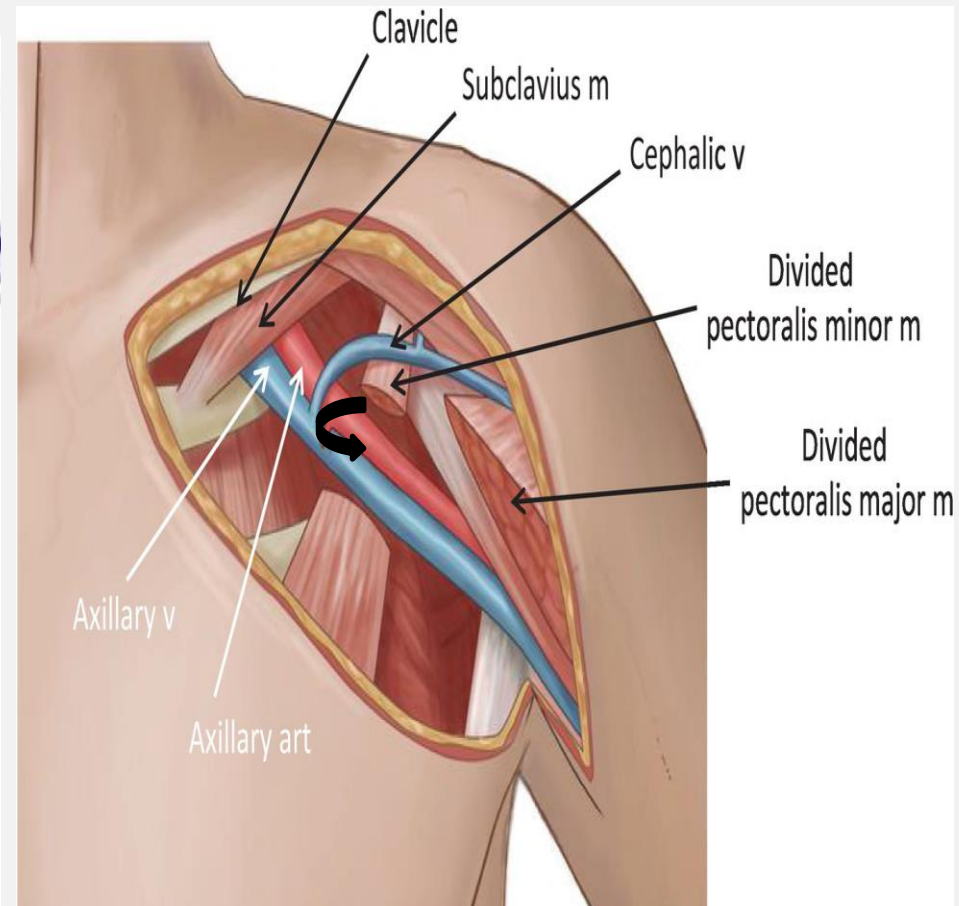
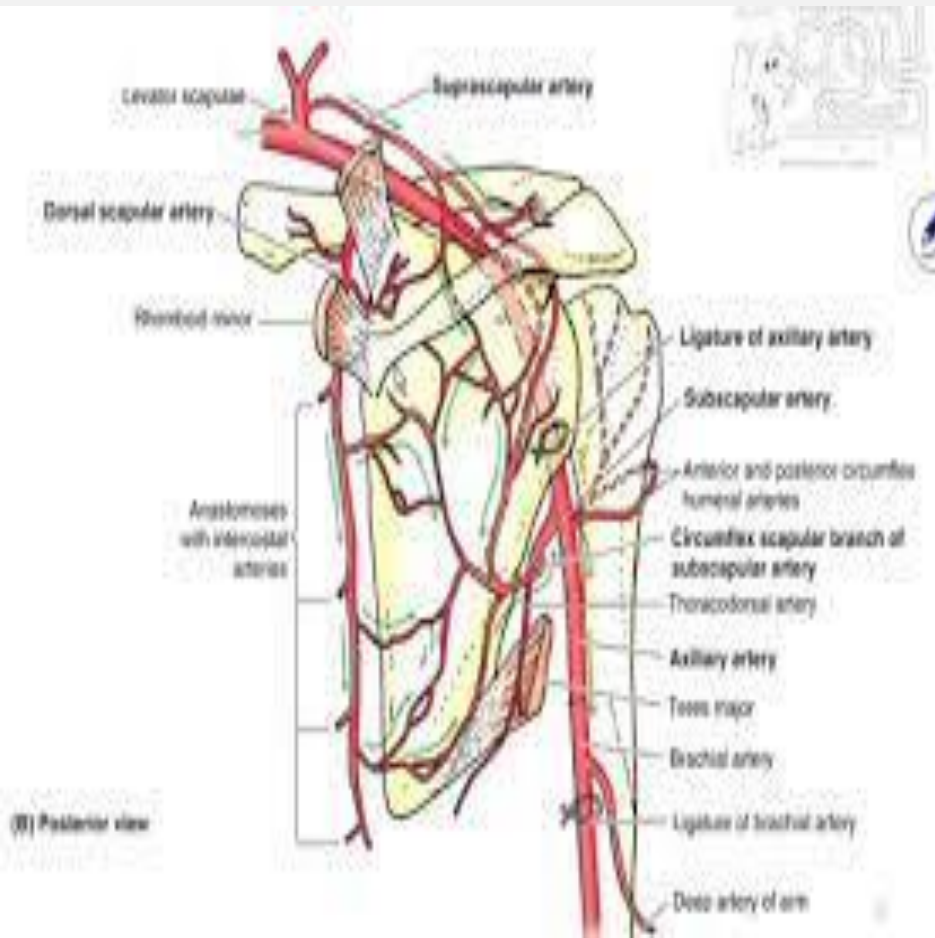
- skin incision is placed along the anterior margin of the deltoid muscle to expose the deltopectoral groove, in which the **cephalic vein** is isolated.
- pectoralis major and deltoid muscles are retracted from each other to expose the coracoid process
- conjoint tendon (pectoralis minor and short head of biceps brachii) attached to its tip divided about two centimeter below the coracoid process to expose the subscapularis tendon,
- which is cut to reach the capsule of the shoulder joint



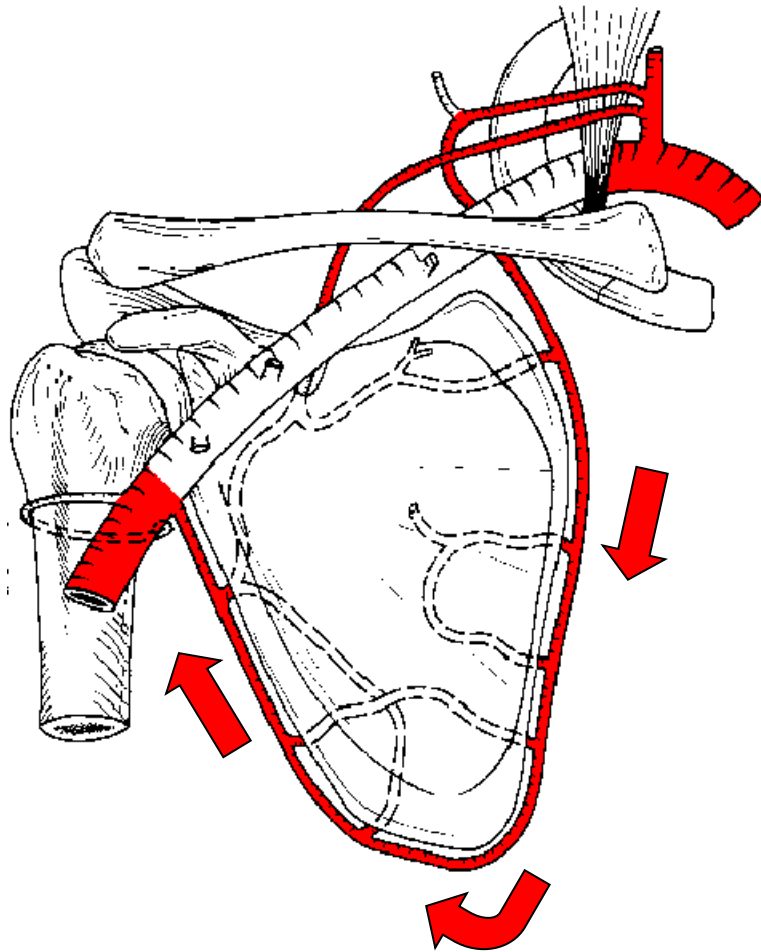


why ligation or blockage of 3rd part of subclavian OR 1st part of axillary arteries do not interfere with blood supply of upper limb?

- Because there is profuse anastomosis (**scapular anastomosis**) between branches of 1st subclavian and 3rd part of axillary arteries provides a collateral circulation which ensures reaching blood to upper limb



Scapular anastomosis



- the axillary artery may be ligated between the thyrocervical trunk and the subscapular artery i.e. between the first part of the subclavian artery and the third part of the axillary artery
- in this case the circulation of blood in the subscapular artery is reversed

□ why the deltoid muscle is preferable site for intramuscular injection?

- Because it is well-developed muscle in most adult & easily accessible
- injection must in lower half of muscle to avoid injury to axillary nerve

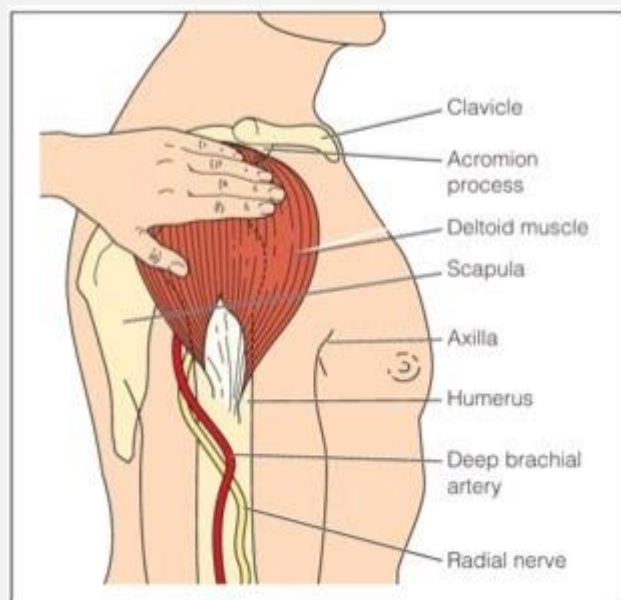
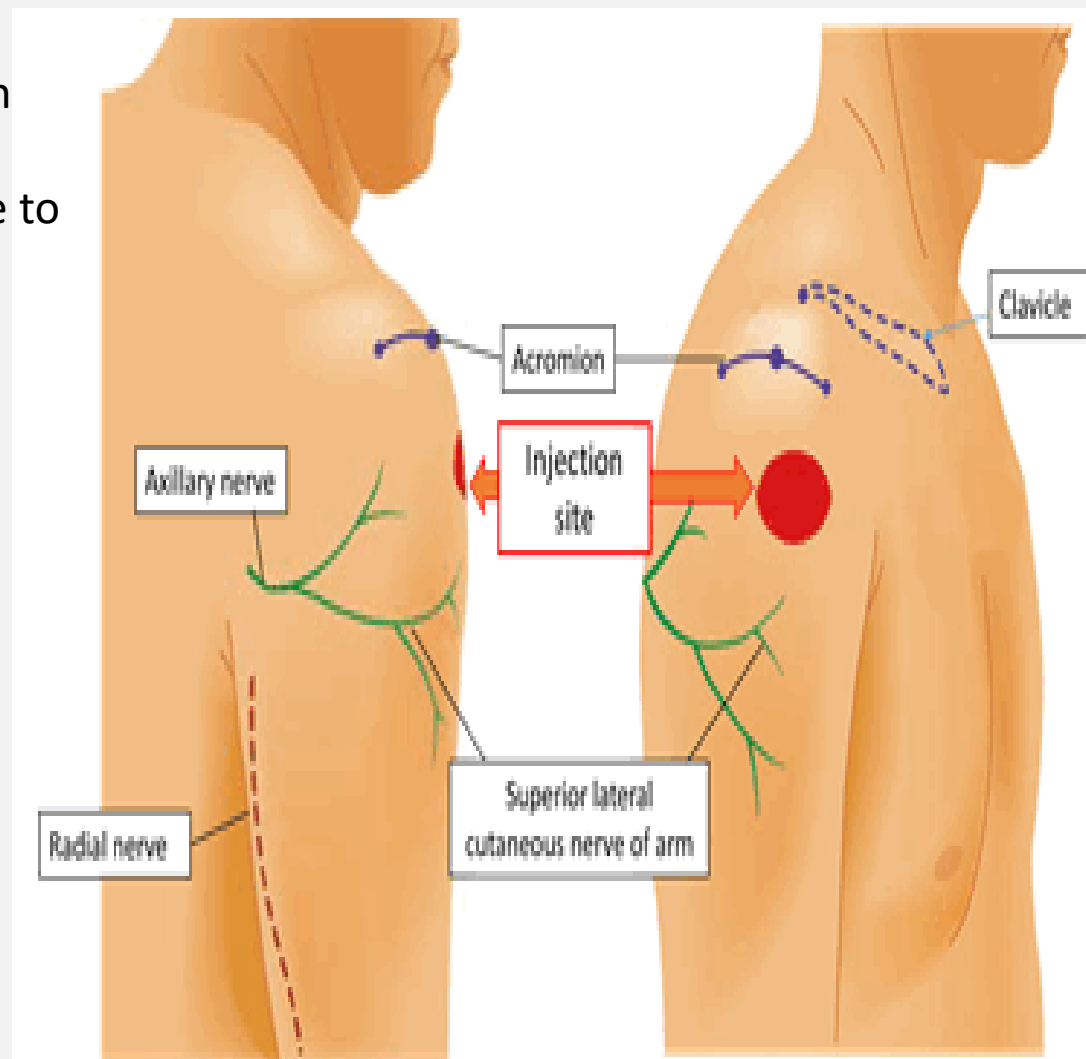


FIGURE 31.38 A method of establishing the deltoid muscle site for an intramuscular injection

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what is the difference between superficial & deep veins?

	Superficial veins	Deep vein
location	located within the subcutaneous tissue	Located deep to deep fascia
Accompany to artery	They not accompany arteries	Usually TWO vein accompany un artery(vena comitantes) & share the name of the artery they accompany
function	Thermal regulation	Drain blood

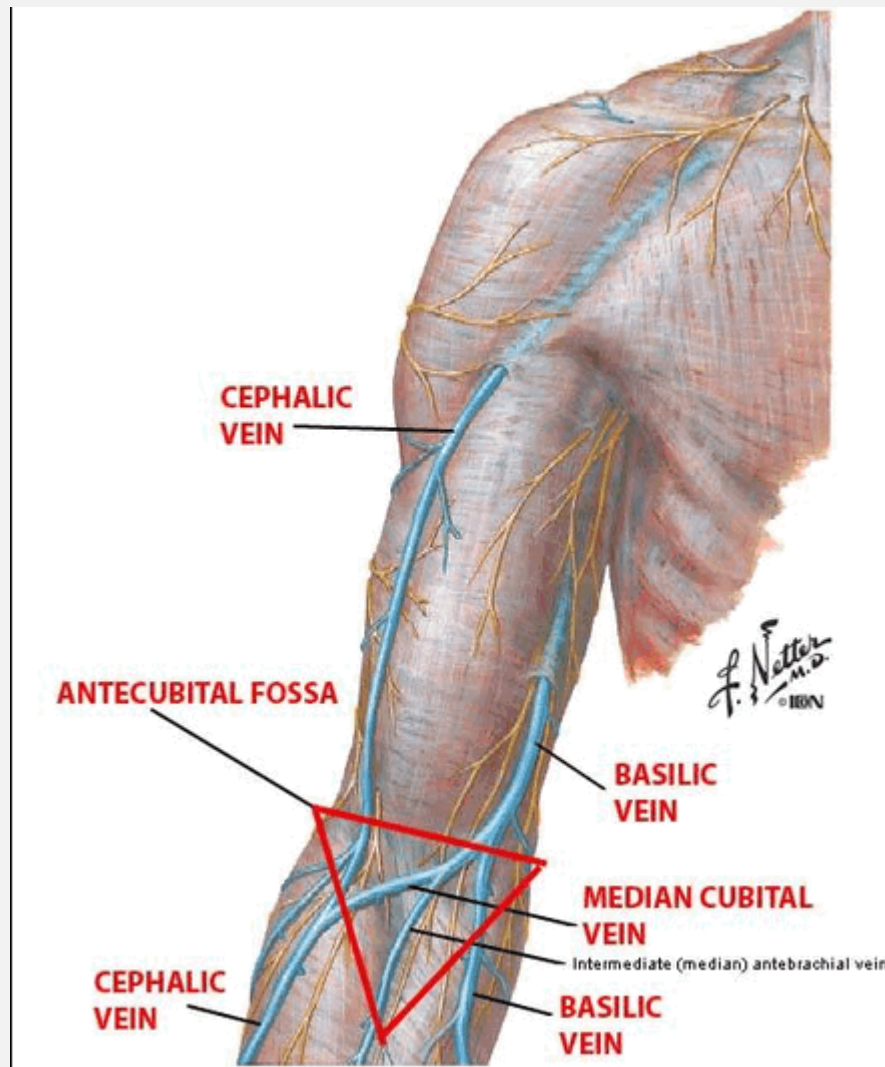
□ why Basilic vein choices for central venous catheterization?

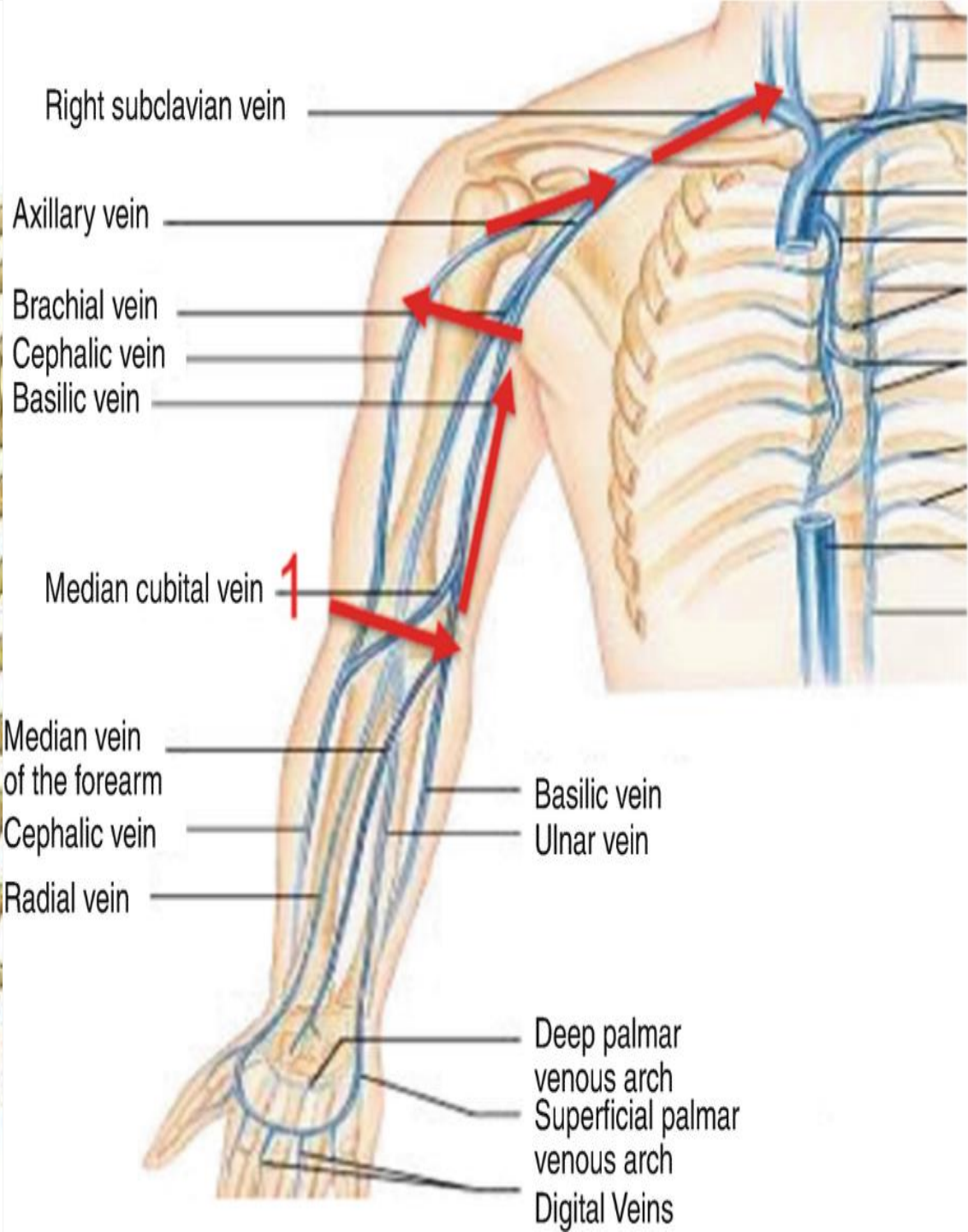
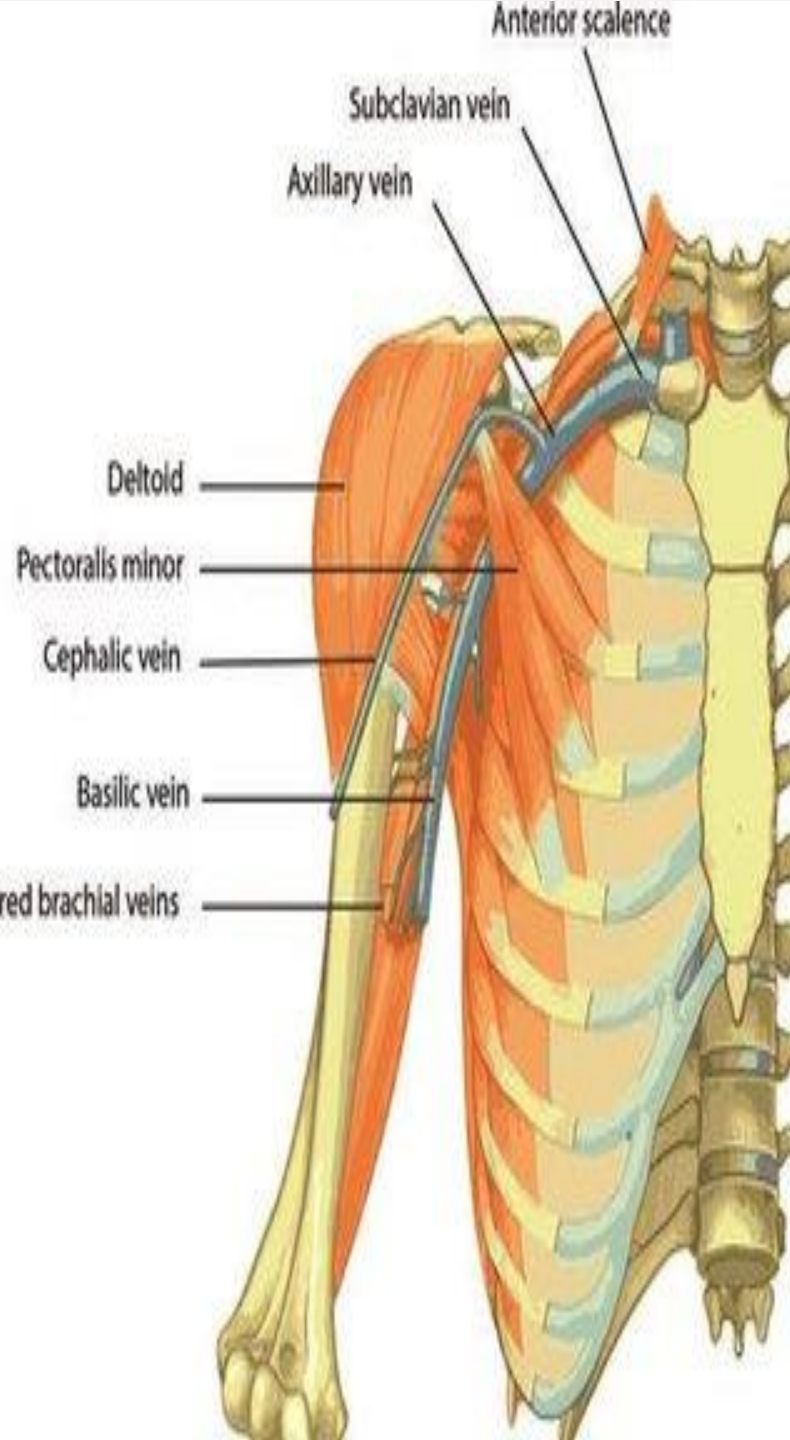
Because

- It increases in diameter as it ascends up from cubital fossa to its termination in axillary vein while cephalic vein doesn't. in contrast, cephalic vein divides into multiple branches as it enters deltopectoral groove
- It is direct in line with axillary vein while cephalic vein terminates to axillary vein at right angle where presence of valve

the catheter passes in succession through the following veins

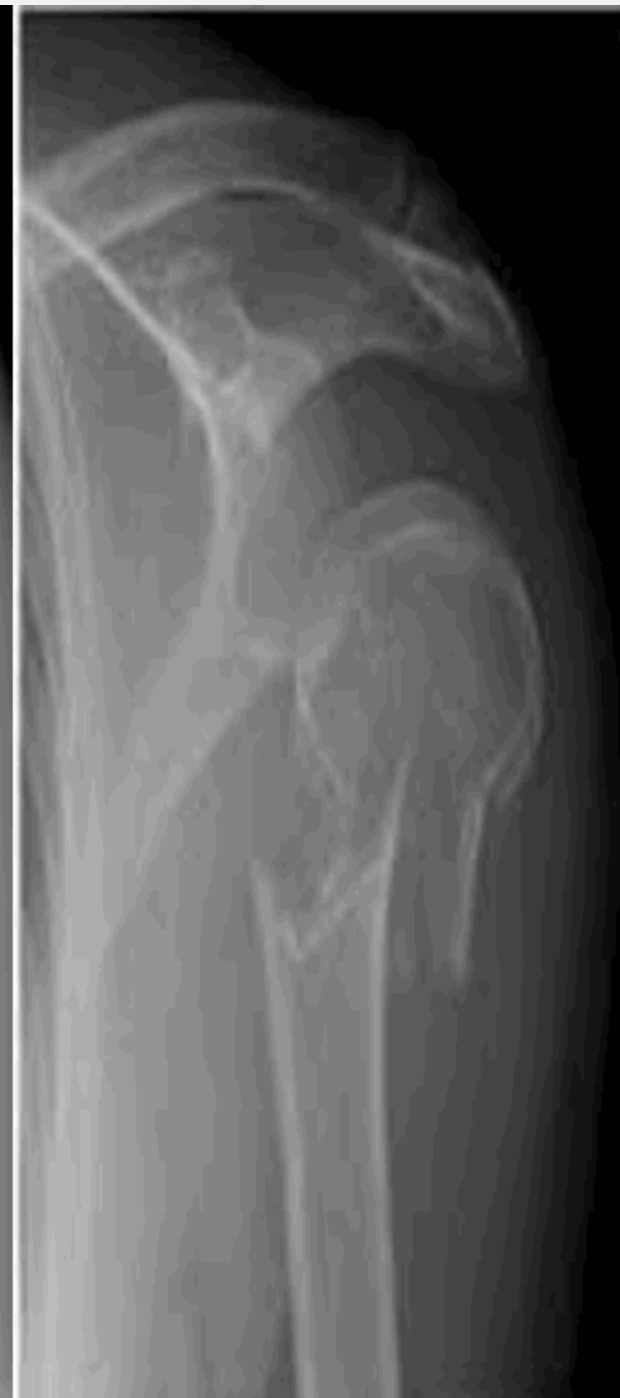
basilic, axillary, subclavian, brachiocephalic & finally the superior vena cava.





 in fractures of humeral shaft, the fractured segments displaced depends on sit of fracture. Explain how is that?

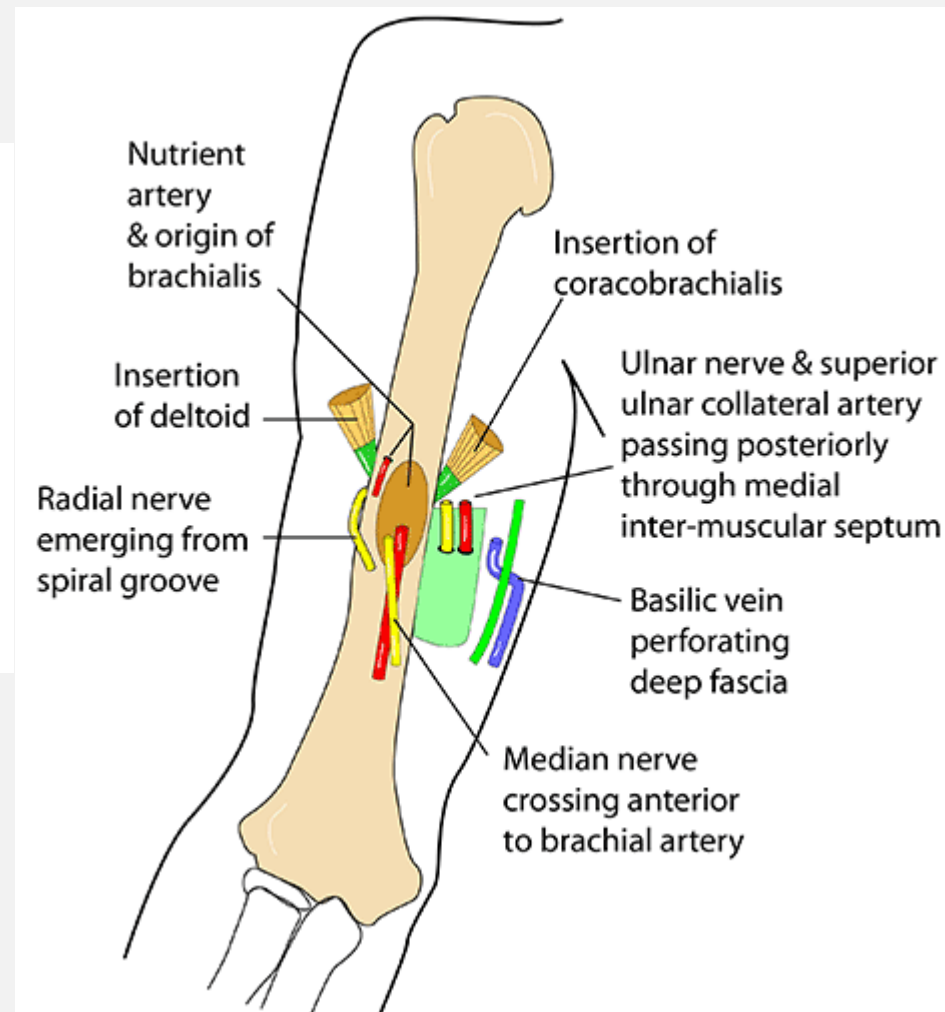
Site of fracture	Proximal segment	Distal segment
Above bicipital groove	Abducted by muscles attached to greater tuberosity(mainly supraspinatus)	Adducted by muscles attached to bicipital groove
Above the level of deltoid tuberosity	Adducted by muscles attached to bicipital groove	Abducted & pulled up by deltoid muscle
Below the level of deltoid tuberosity	Abducted by deltoid	Adducted & pulled up by biceps & triceps





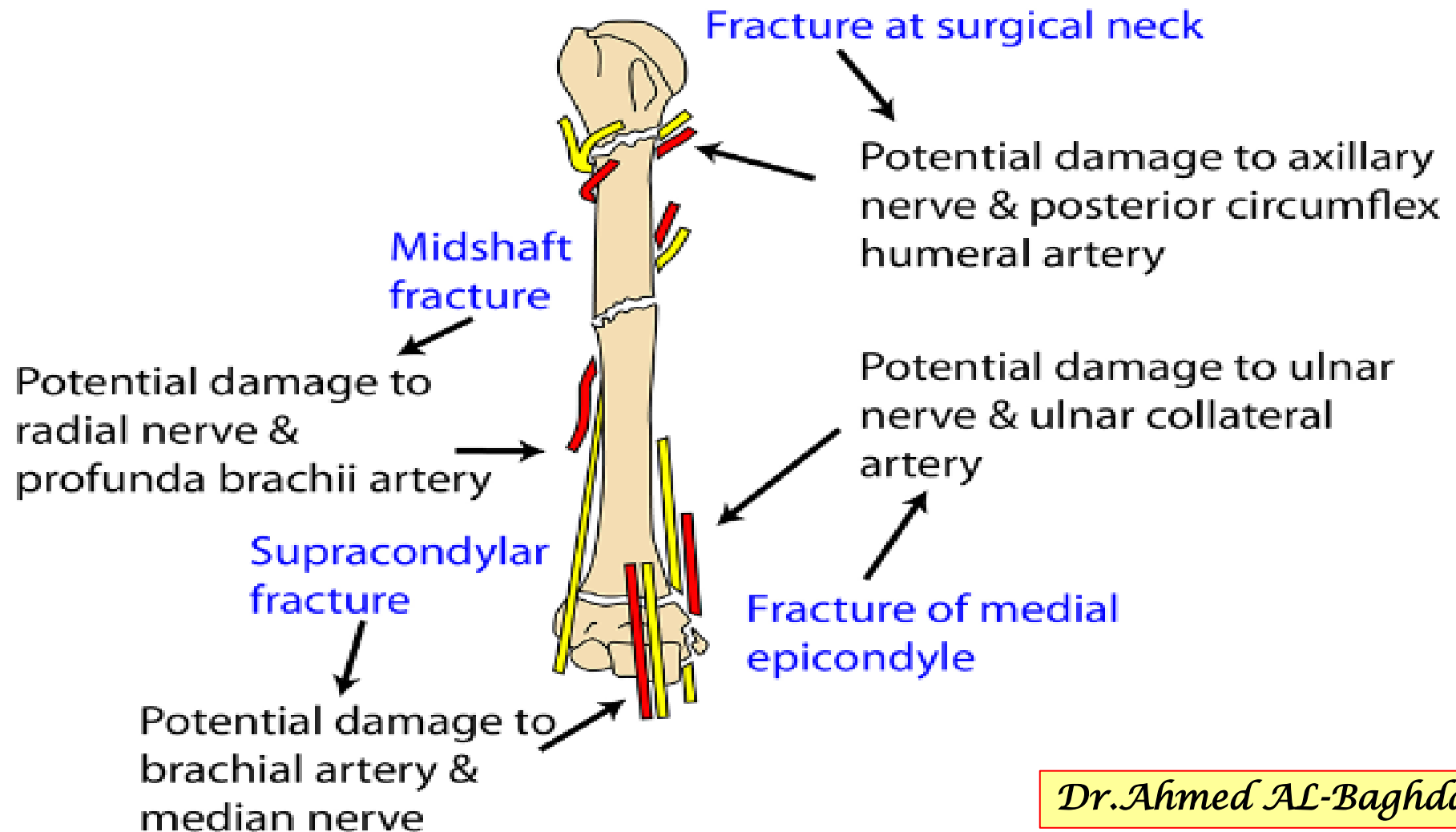
what are the anatomical events that occur at or near mid-shaft of humerus?

- Nutrient artery entering humerus
- Origin of brachialis
- Insertion of deltoid
- Insertion of coracobrachialis
- Radial nerve emerging from spiral groove
- Basilic vein perforating deep fascia
- Median nerve crossing anterior to brachial artery
- Ulnar nerve & superior ulnar collateral artery passing posteriorly through medial inter-muscular septum



❑ What are the structures in direct contact with humerus and liable for injury?

ARTERIES AND NERVES LIABLE TO DAMAGE AT 4 SITES OF FRACTURE OF THE HUMERUS





❑ What is Volkmann's ischemic contracture?

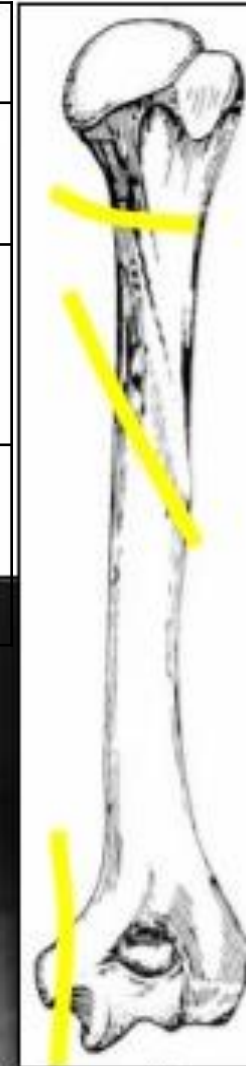
In supracondylar fracture

- There is displacement of proximal bone fragment forward & downward, compressing brachial a. & median n.
- When the collateral circulation cannot compensate for occlusion of the brachial artery leads to ischemia of flexor muscles of forearm & necrosis
- the muscles replaced by fibrous tissue leading to thinning & shortening of the muscle
- result in flexion deformity of wrist & fingers **(Volkmann's contracture)**



Fractures of the humerus

nerve	Site of contact	deformity
Axillary	Around surgical neck	Flat shoulder
Radial	Spiral groove	Wrist drop
ulnar	Behind medial epicondyle	Claw hand

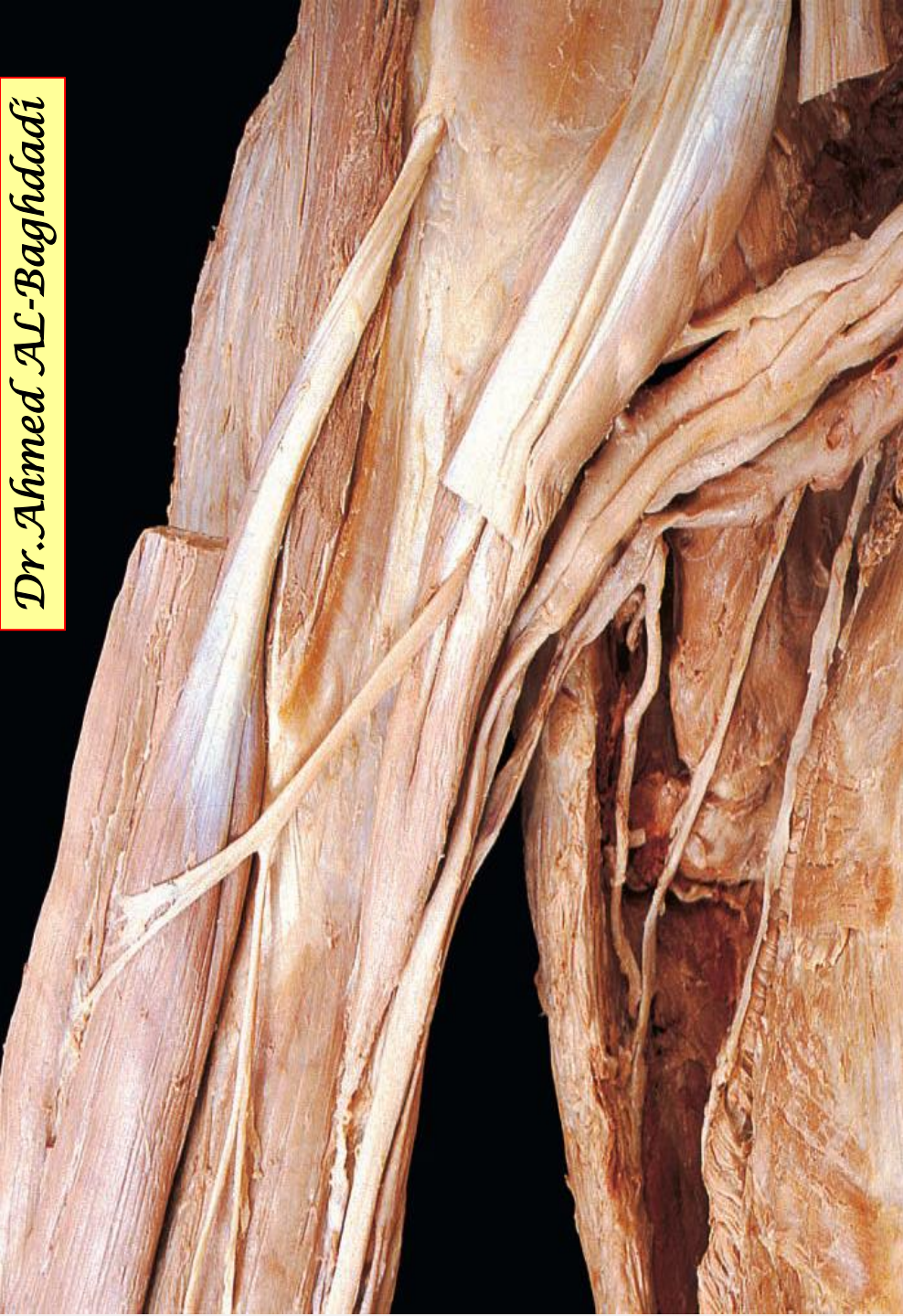


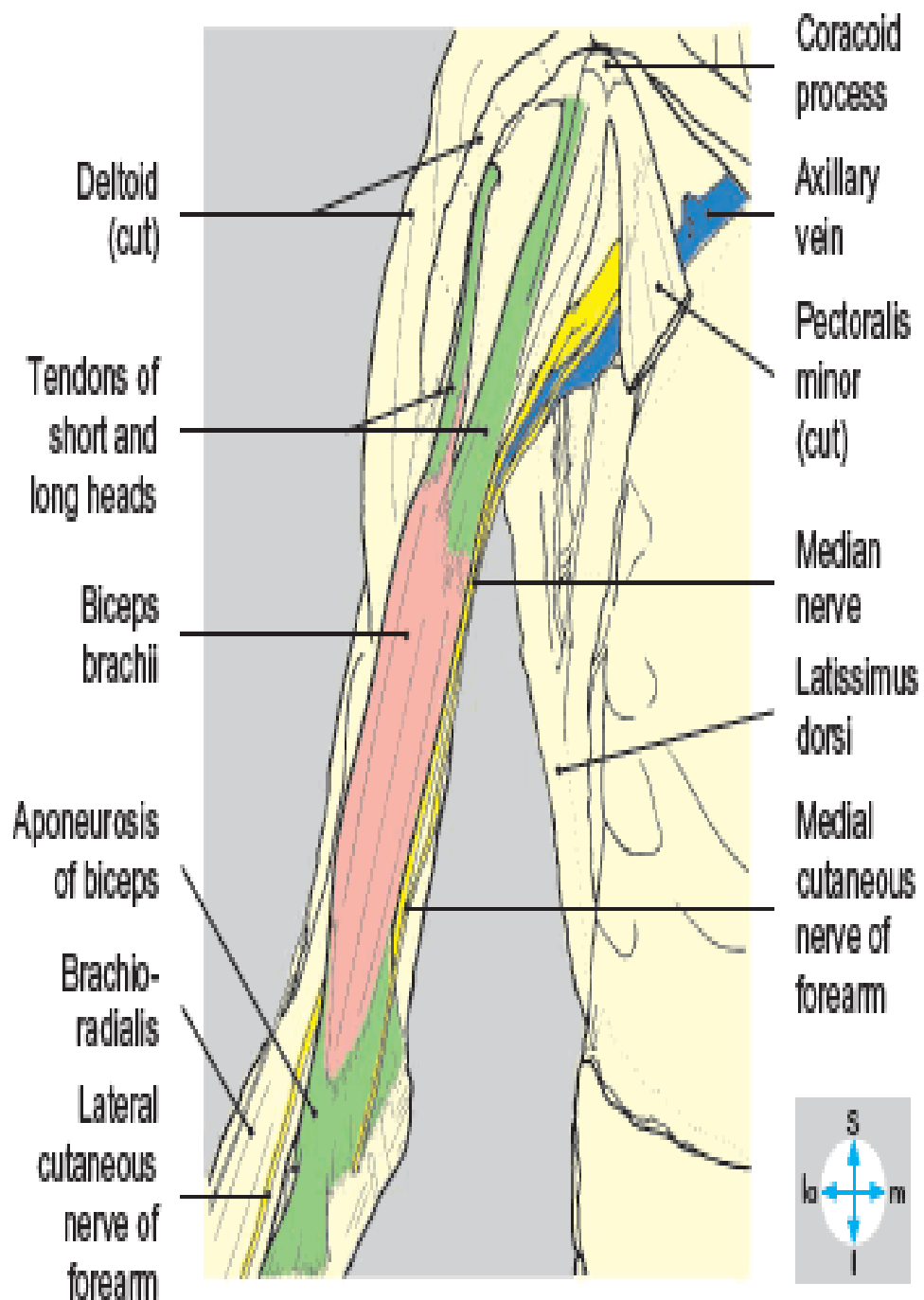
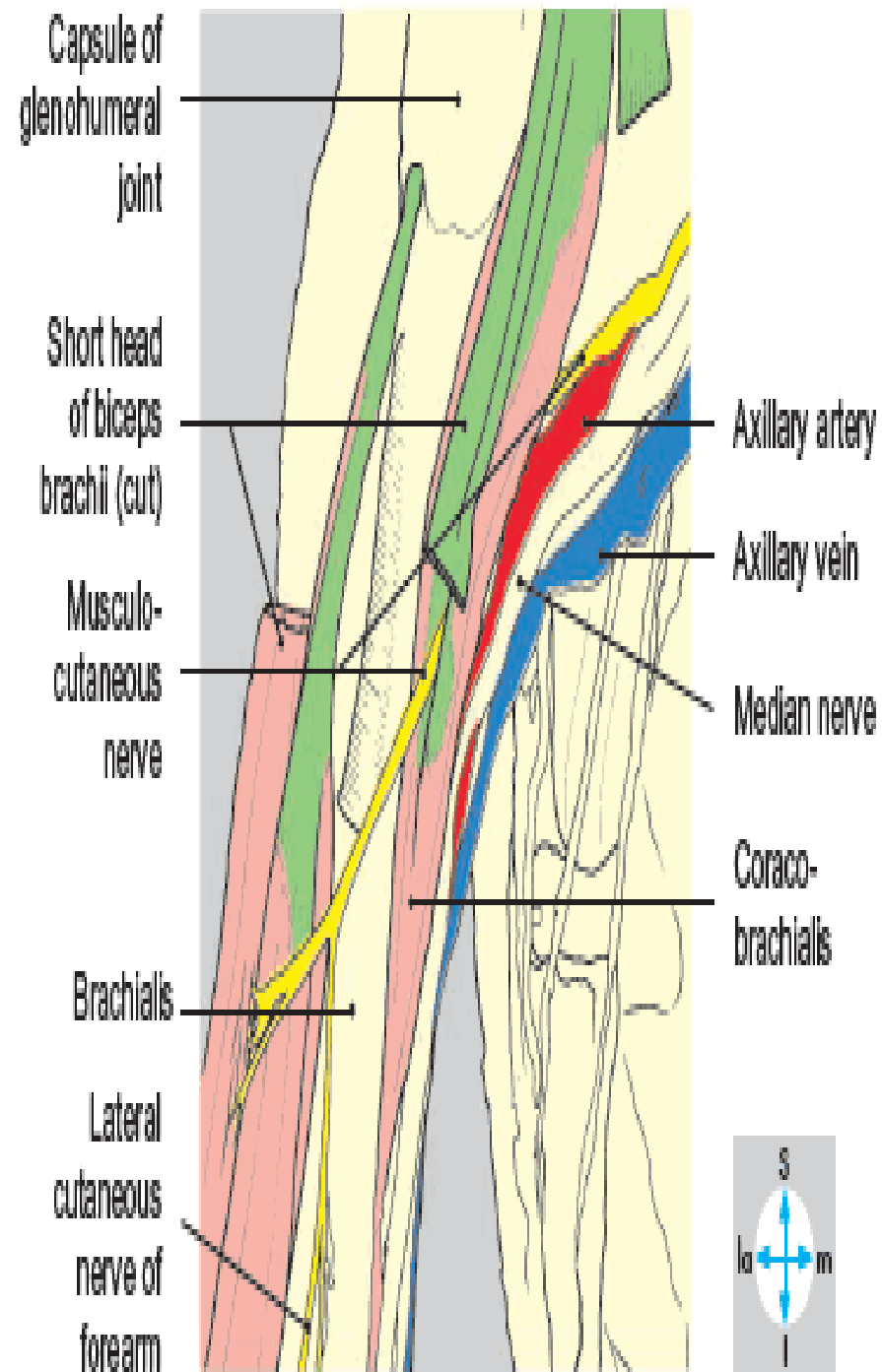
Surgical neck fracture



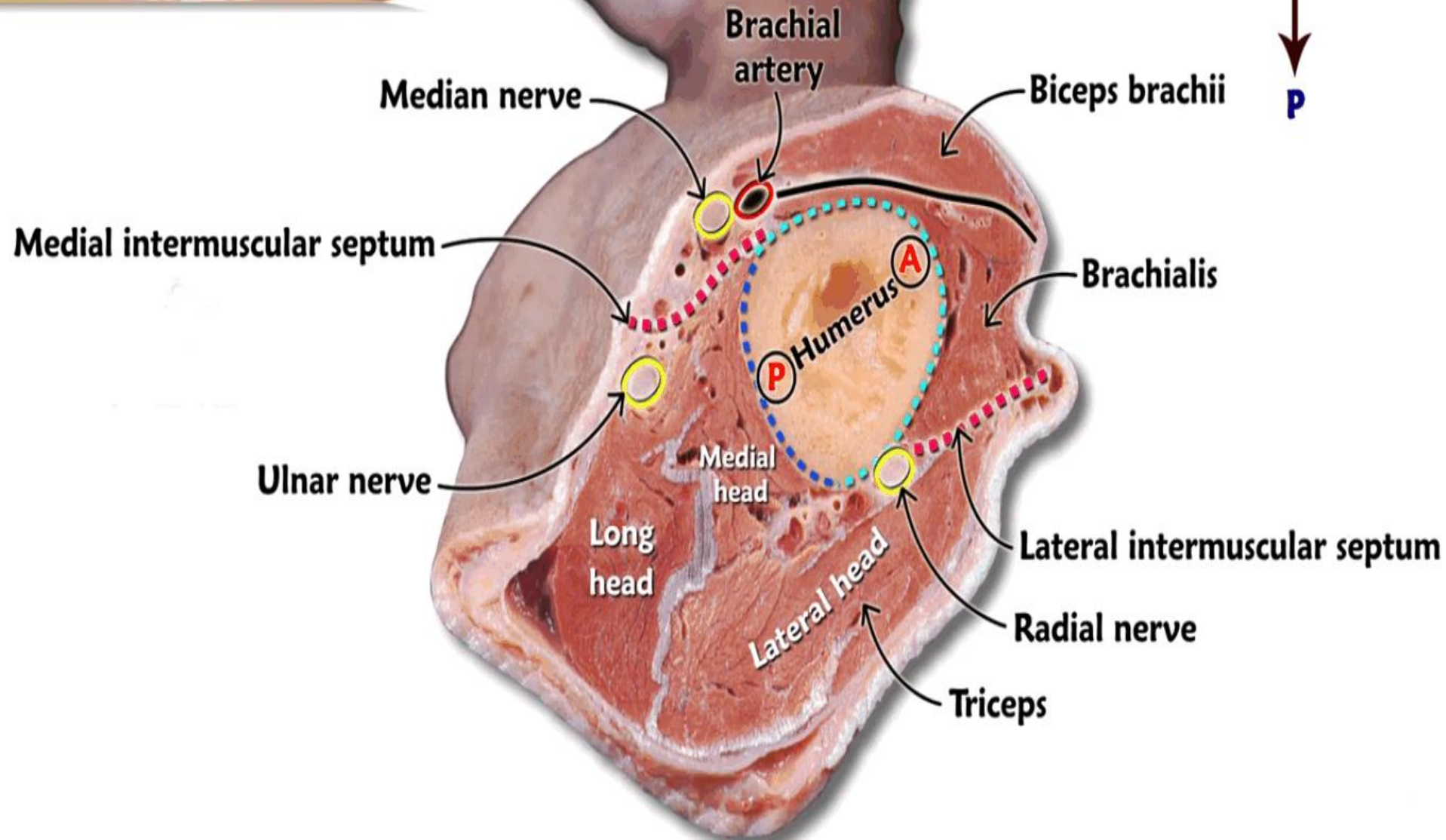
Medial epicondyle fracture

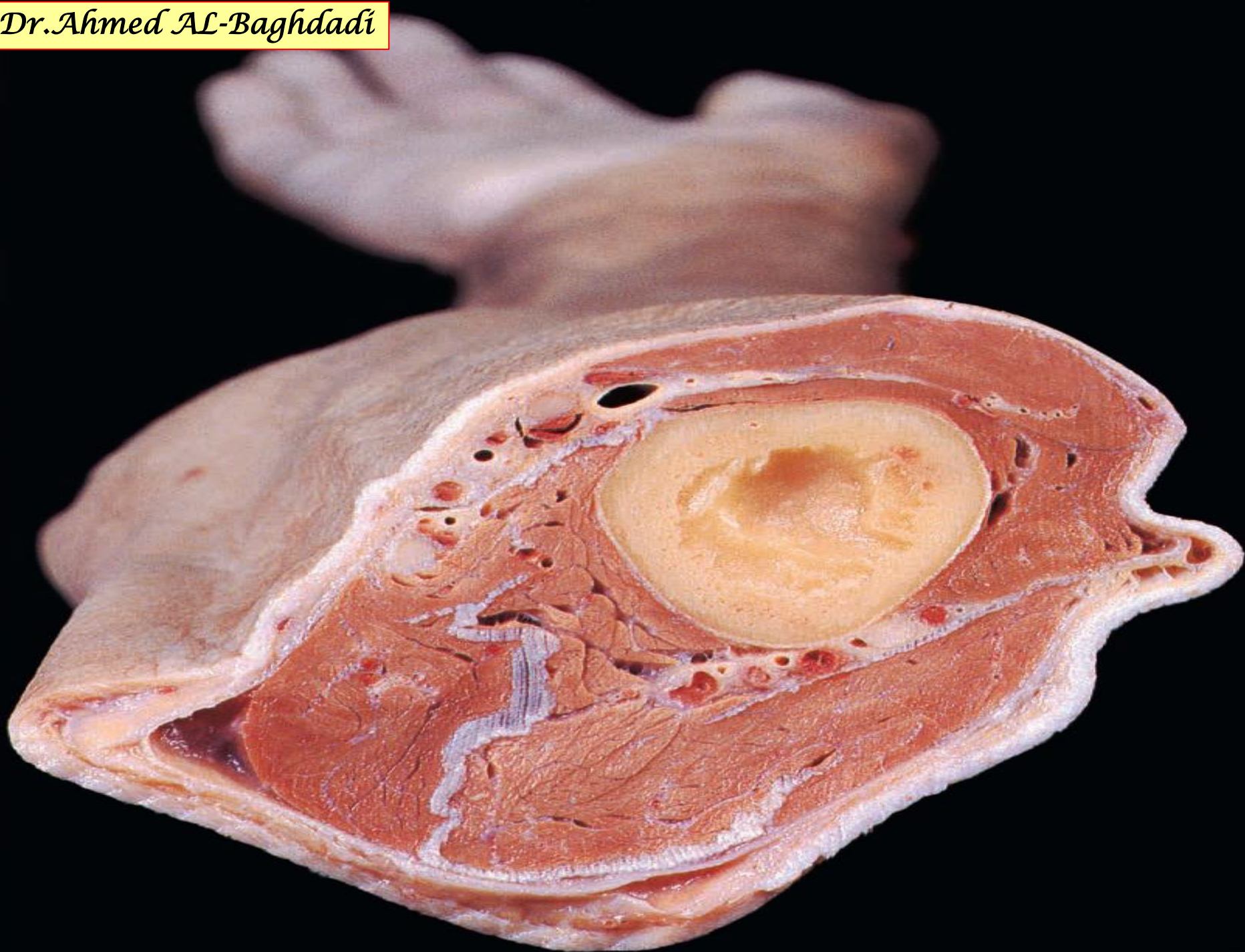
Spiral groove fracture

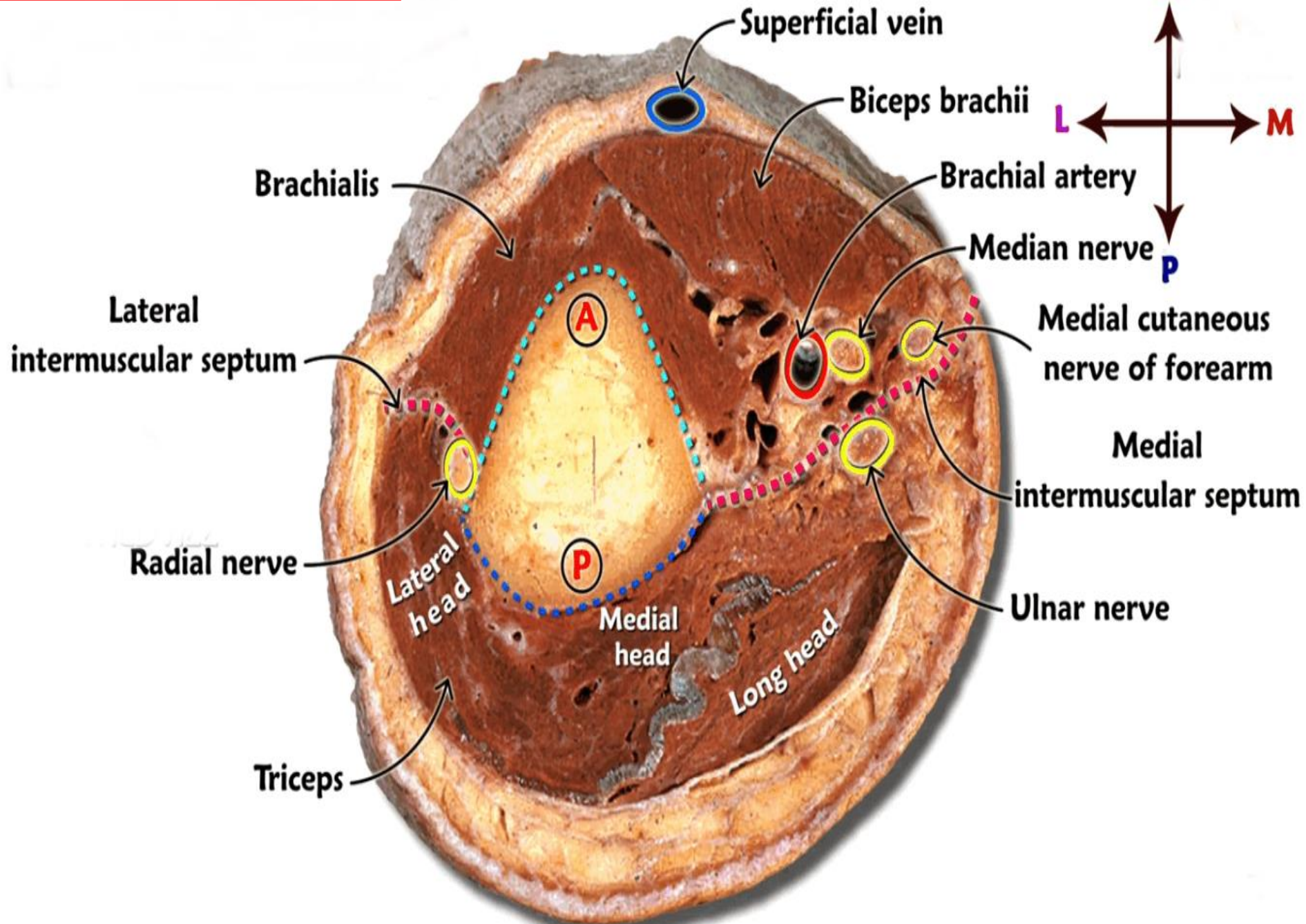


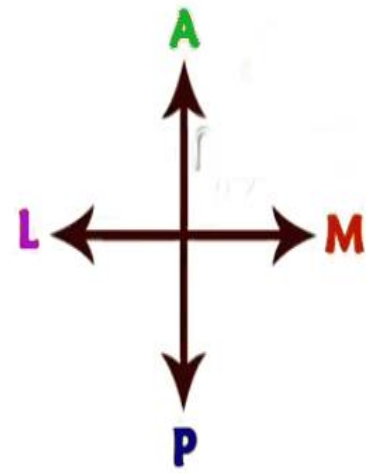


Posterior compartment









❑ one is able to extend his elbow, if radial nerve is injured in radial groove. explain?

- Because long & medial heads of triceps (chief extensor of elbow) are supplied by radial nerve in axilla(before entering radial groove





Superior
Lateral

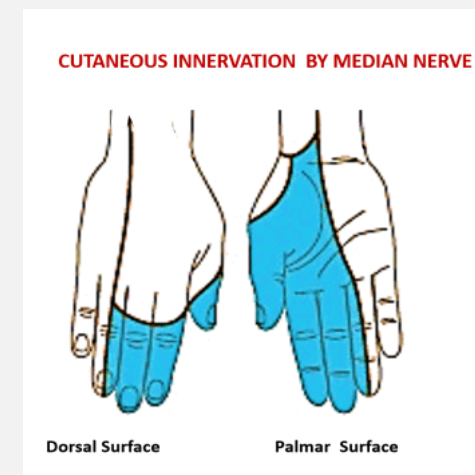
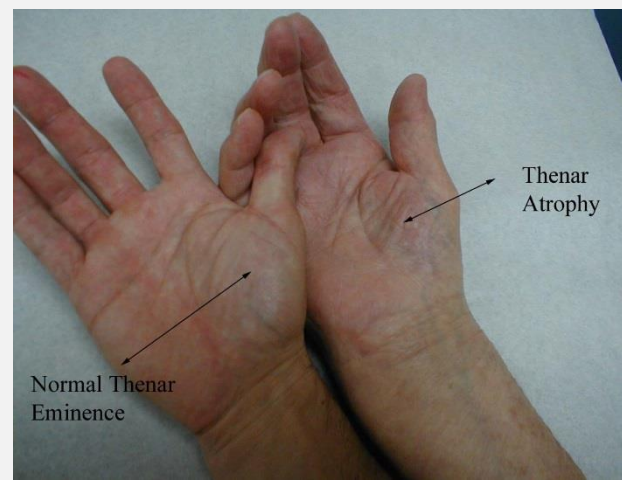
what is the effect of median nerve injury at elbow?

Motor deficit

- Loss of pronation
- Weakened flexion at wrist
- Ape hand deformity
- Benediction hand: when patient tries to make fist, the index & middle fingers remain extended due to paralysis of flexors of these digits

▪ Sensory deficit

loss over lateral half of palm & lateral 3 1/2 digits





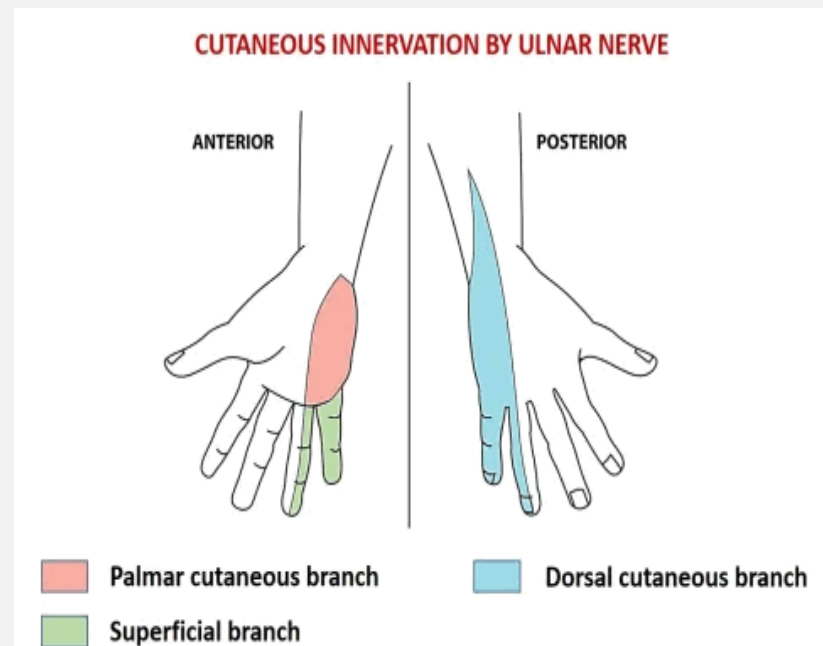
what is the effect of ulnar nerve injury at elbow?

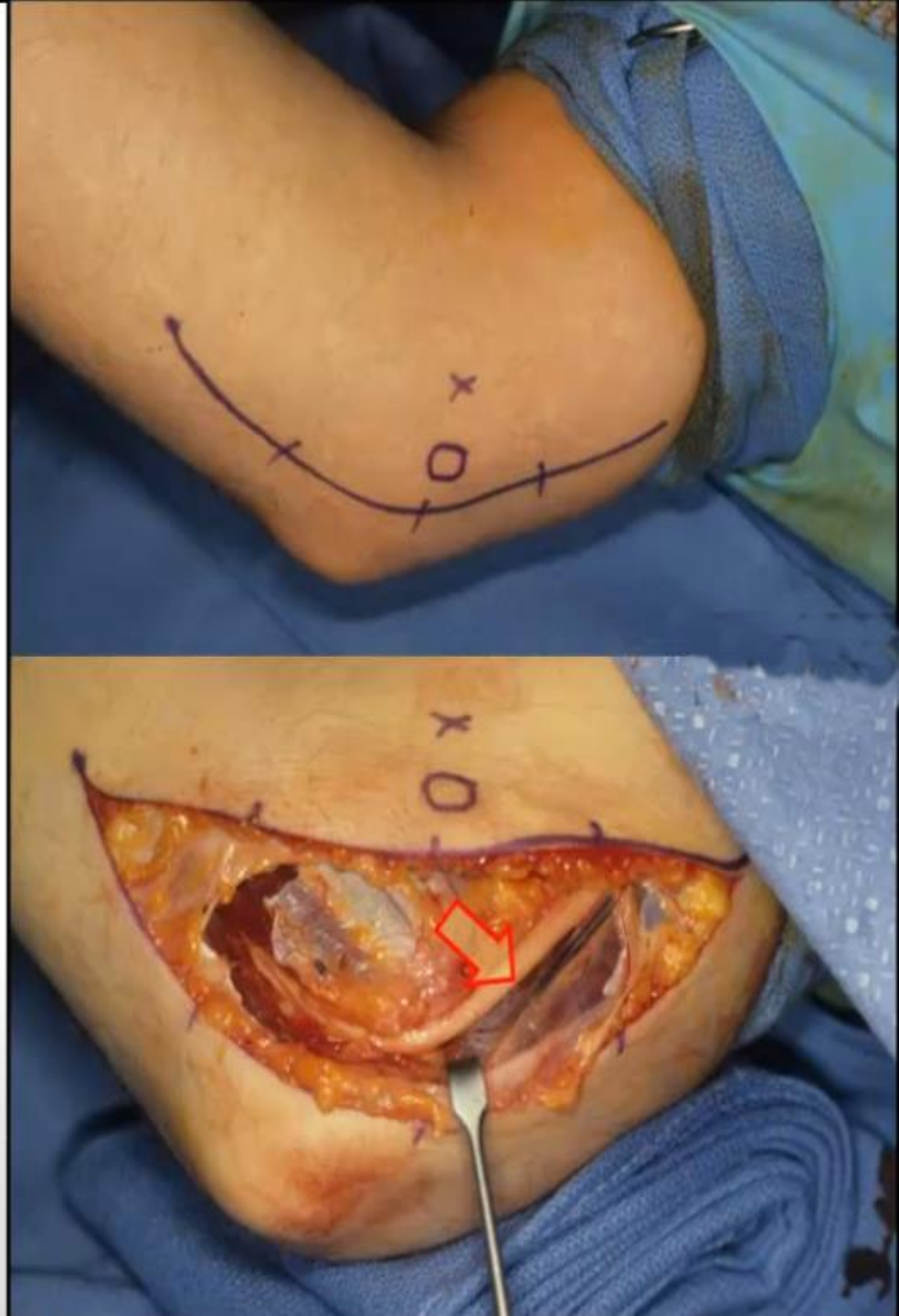
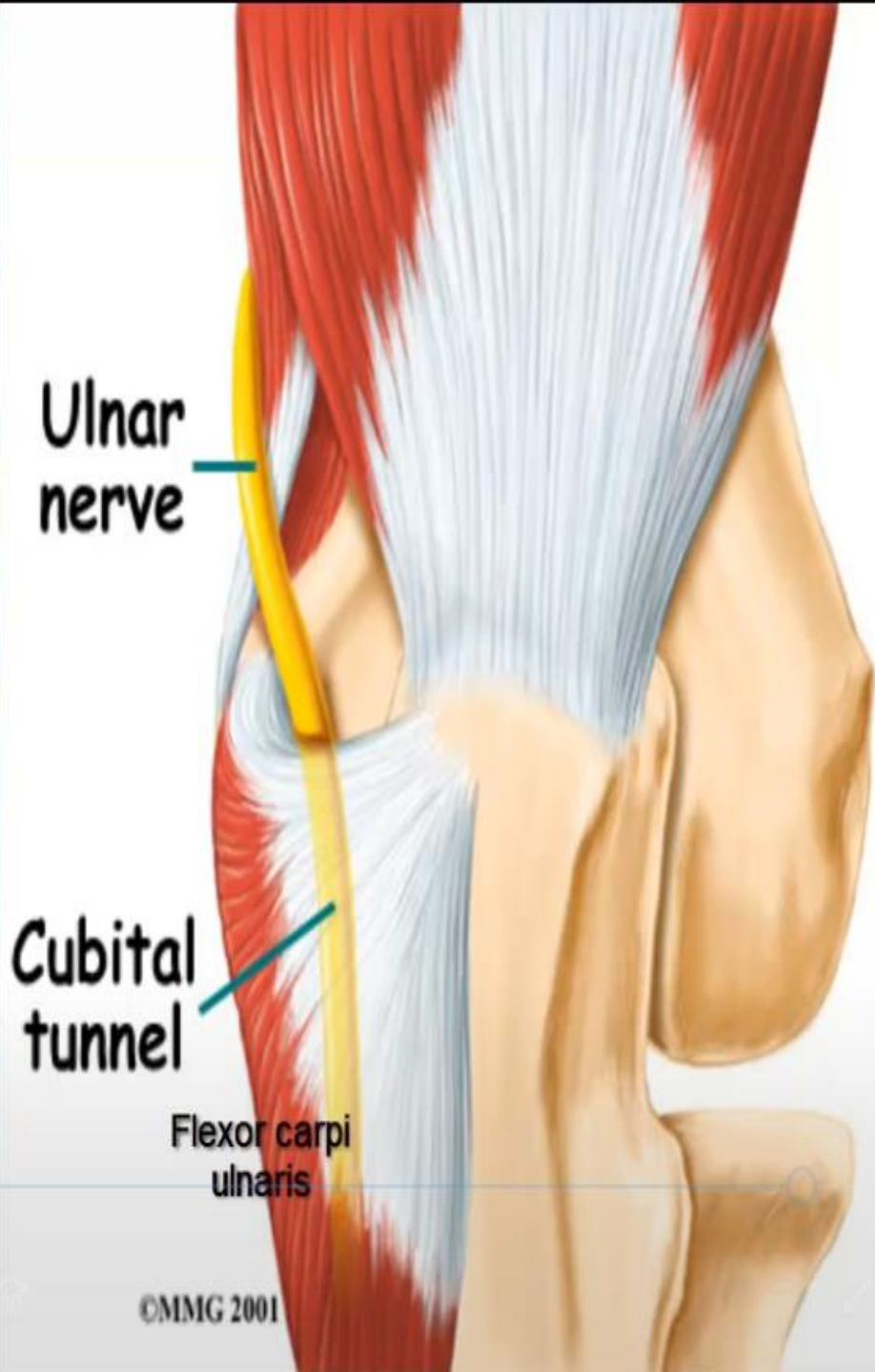
Motor deficit

- Partial claw hand affects little & ring fingers: hyperextension at their MP joints & flexion at their IP joints due to paralysis intrinsic mm. of hand
- 1st & 2nd lumbricals supplied by median nerve therefore they spared
- Flattening of Hypothenar eminence
- Loss adduction & abduction of fingers due to paralysis of dorsal & palmar interossei

Sensory deficit

loss over palmar & dorsal surfaces of medial 1/3rd of hand



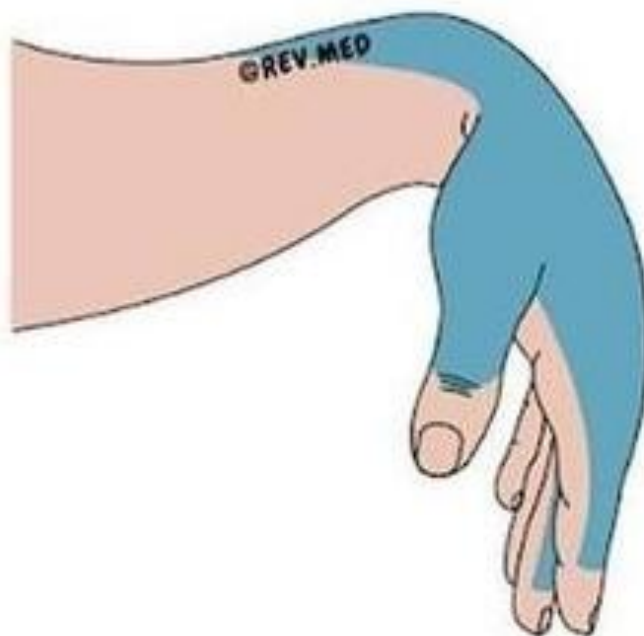


**HAND OF
BENEDICTION**



**Median Nerve
Injury**

**WRIST
DROP**



**Radial Nerve
Injury**

**CLAW
HAND**



**Ulnar Nerve
Injury**



why anatomy of cubital fossa clinically important?

- 1- it is a common site for venipuncture
- 2- It is site for recording blood pressure
- 3- Its contents are liable to injury in supracondylar fracture of humerus



Anterior view of Radial N
(between Brachialis
and
Brachioradialis)

Superior
Lateral

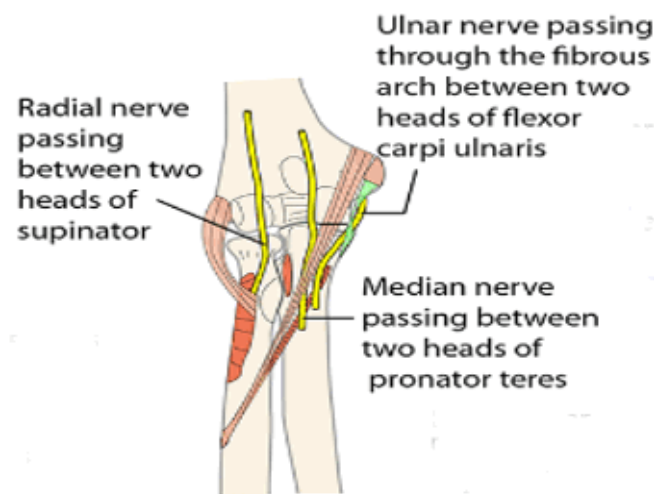
A photograph of a dissected human arm showing the radial nerve. The nerve is a thick, yellowish, cord-like structure running along the humerus. It is surrounded by a network of smaller, branching nerves and blood vessels. The surrounding muscles, including the brachialis and brachioradialis, are visible as reddish-brown, fibrous structures. The dissection is performed on a light-colored, textured surface. In the bottom right corner, there is a diagram with two arrows: one pointing upwards labeled 'Superior' and one pointing to the left labeled 'Lateral'.

Cubital Fossa

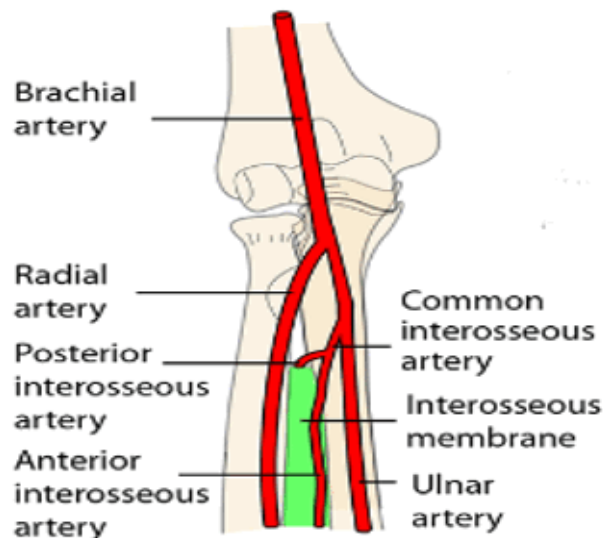


Relations of nerves & arteries in the elbow region

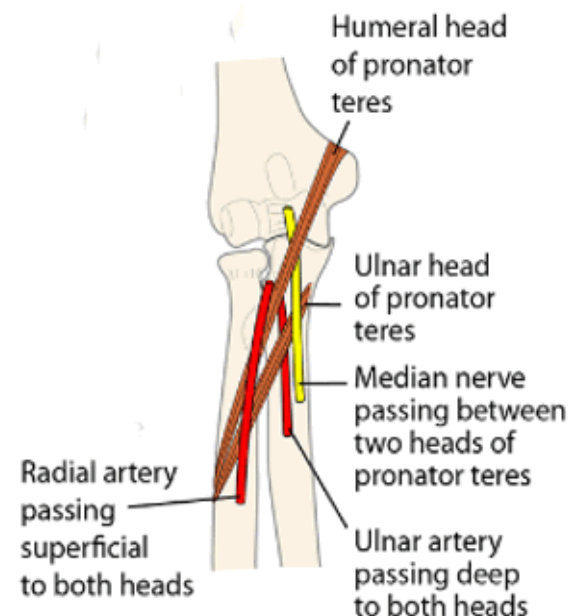
- 1- **median n.:** leaves the fossa by passing between the 2 heads of pronator teres
- 2- **the terminal part of brachial a.** & the beginning of both radial & ulnar arteries
 - a) radial a. leaves the fossa by passing superficial to pronator teres
 - b) ulnar a. leaves the fossa by passing deep to pronator teres m.
- 3- **biceps tendon** passes backwards to reach its insertion in the radial tuberosity
- 4- **radial n. & its post. Interosseous br.**
 - **Radial n.** leaves the fossa by passing deep to the brachioradialis m.
 - **Post. Interosseous n.** leaves the fossa by piercing the supinator m.



Nerves never pass between bones but pass around them. Radial, median and ulnar nerves all pass between the two heads of muscles



Arteries may pass between two bones as here where the posterior interosseous artery passes between ulna and radius



Relations of the two heads of pronator teres



why auscultation of brachial artery is commonly used to record arterial blood pressure?

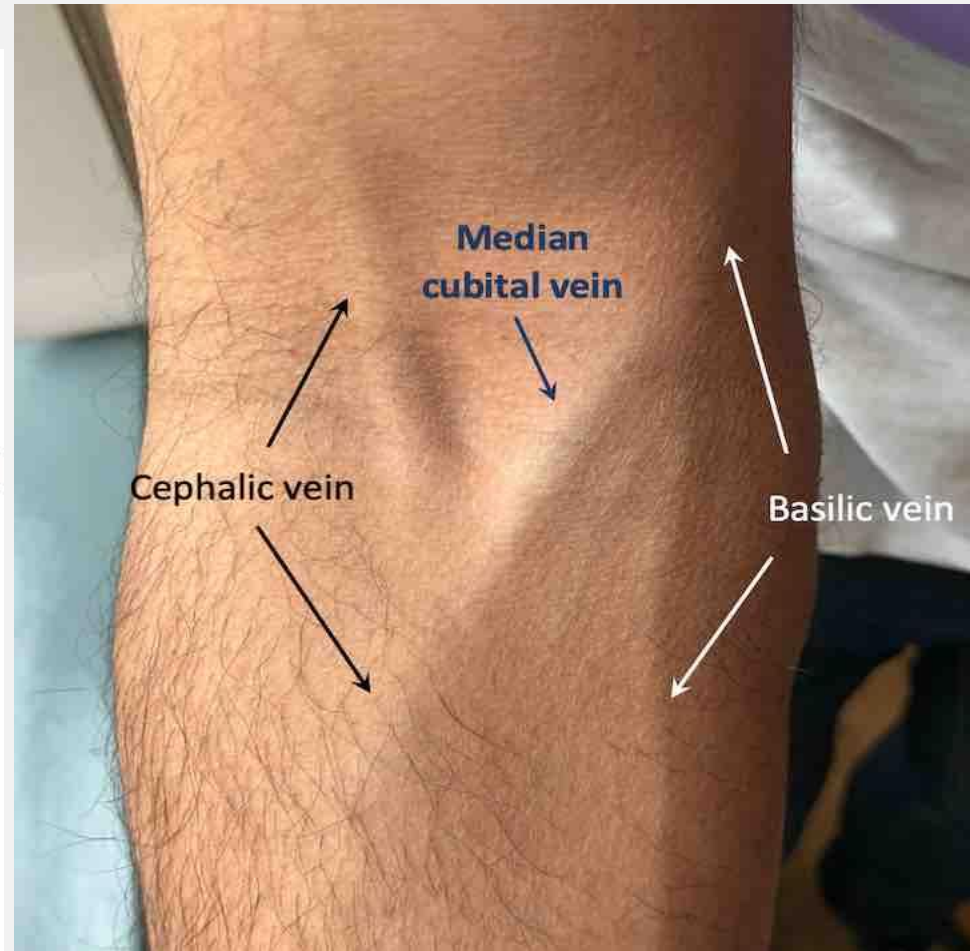
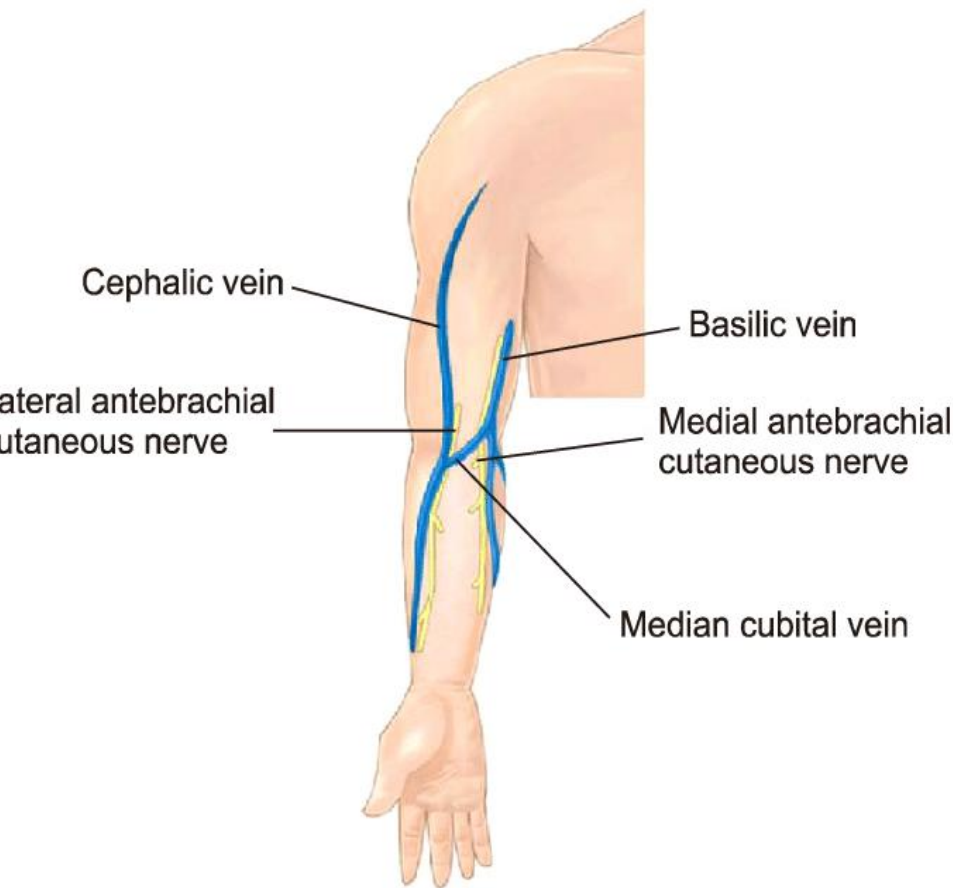
- 1- it is superficial , infront of elbow and its pulsations are easily felt just medial to tendon of biceps m.
- 2- triangle hollow infront elbow fossa provides a flat area to place the stethoscope over artery

Dr. Ahmed AL-Baghdadi



❑ why the elbow must be kept extended during venipuncture of median cubital vein?

- ❑ It lies superficial to bicipital aponeurosis which separates it from median nerve and brachial artery
- ❑ When elbow extended , the aponeurosis become tense and provides a firm barrier protecting artery and nerve





what is the difference between medial & lateral epicondyles?

pronation

supination

1) It is initiated by brachioradialis (rotates the radius from full **PRONATION** to midprone position.

2) It is completed by

- supinator (when elbow is extended)
- Biceps (when elbow is flexed)

3) antigravity movement(**screwing**) so it done by **more powerful muscles**

4) Radial nerve

1) It is initiated by brachioradialis (rotates the radius from full **SUPINATION** to midprone position.

2) It is completed by

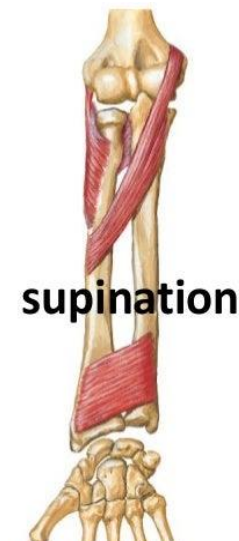
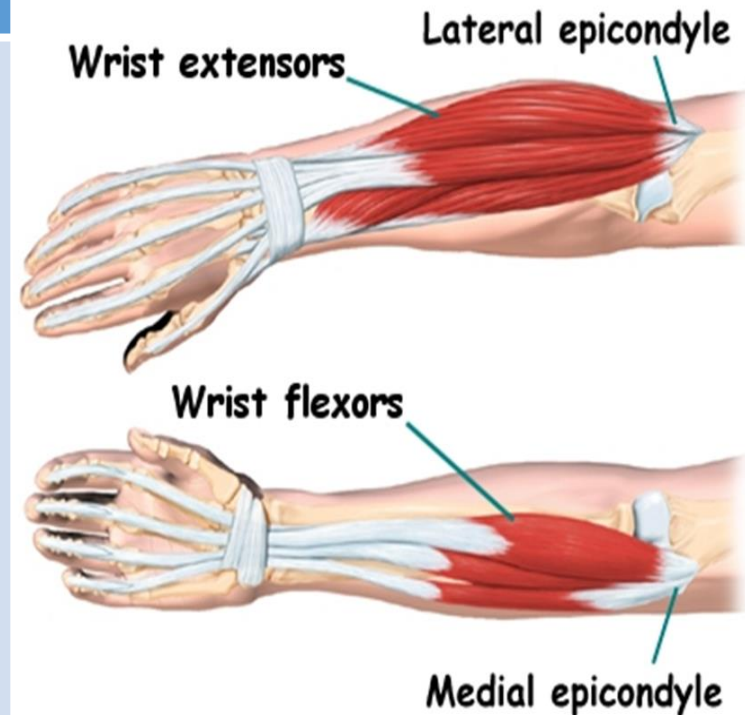
- Pronator teres
- pronator quadratus

3) gravity helps movement of pronation(**unscrewing**)

4) Median nerve

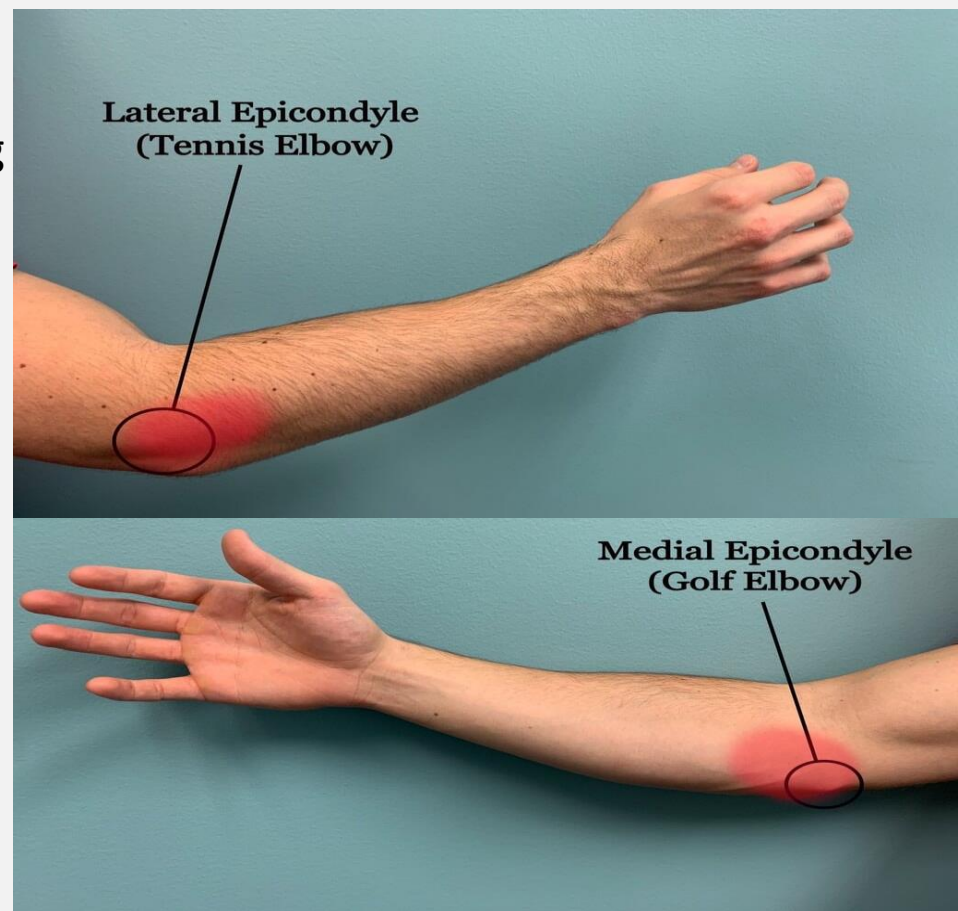
- A lesion of median nerve at or above elbow lead to loss of ability to pronate the forearm.
- radial nerve lesion, no loss supination

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what are tennis & golfer's elbow?

- Medial & lateral **epicondylitis**
- Elbow problems occur where the tendon attaches to bone
- These problems occur due to repetitive stress which causes **interstitial tearing**
- Causing inflammation, soreness, swelling & pain
- **Lateral epicondylitis**: in sports which required twisting & extension of wrist against resistance e.g. tennis
- **Medial epicondylitis**: in sports which required repetitive overuse activities which aggravated by movement of wrist





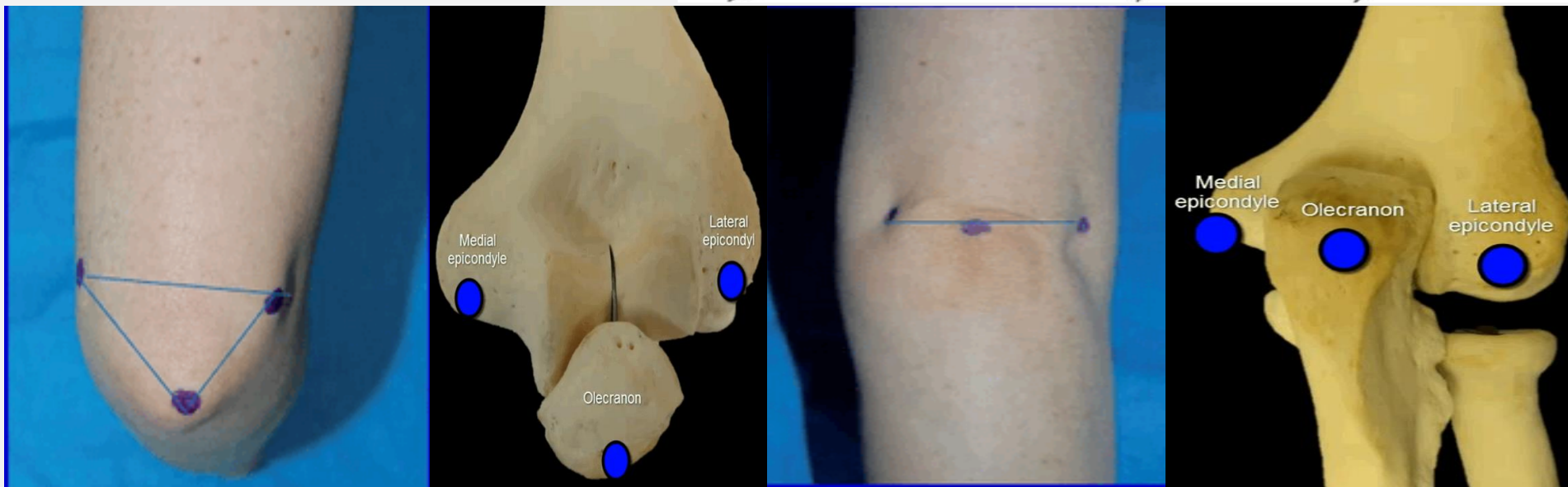
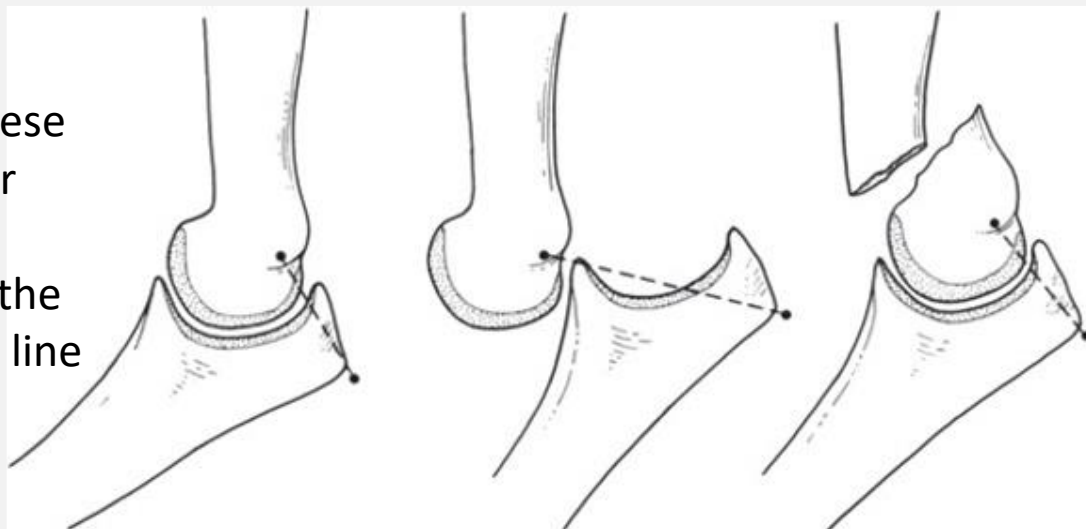
what Popeye elbow?

- Elbow bursitis: (inflamed olecranon bursa) filled sac at the tip of elbow due to trauma, infection, or inflammation
- **Elbow bursa**: provides protection & decreases friction to elbow joint
- Causes pain, swelling, tenderness and lump in area at the tip of elbow
- May be very difficult to put the elbow down on a surface due to the tenderness



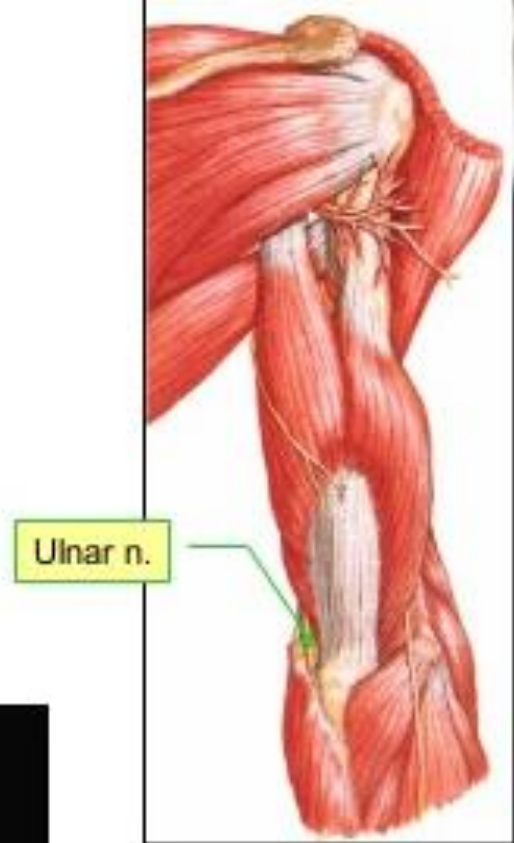
□ why in supracondylar fracture of humerus ,triangle relationship of the 3 bony points(olecranon, medial & lateral epicondyles) is not disturbed but in elbow dislocation , it is disturbed?

- A supracondylar fracture lies above these points, which therefore remain in their triangular relationship to each other
- in dislocation of the elbow, however, the olecranon comes to lie more or less in line with the epicondyles



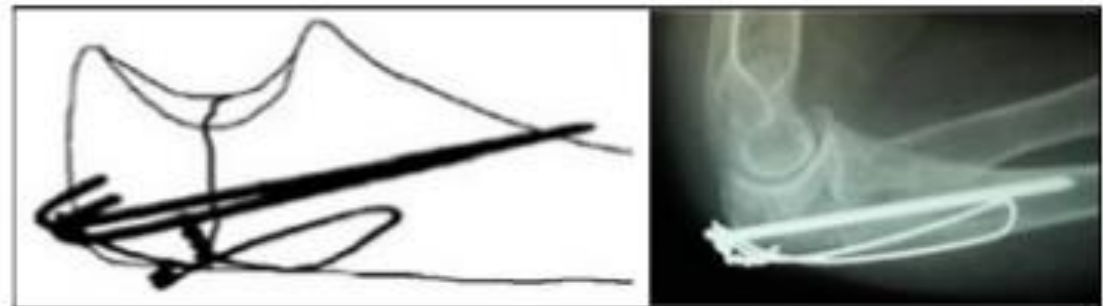
Dislocation of the elbow

- Commonly dislocates posteriorly by an indirect force such as a fall on the hand.
- **Anatomical complications:**
- Ulnar nerve injury.
- Injury of the brachial artery.



Fracture of olecranon

- Direct violence or avulsion by forcible contraction of triceps.
- The fragments are widely separated and require surgical repair to reconstruct the integrity of the elbow joint.



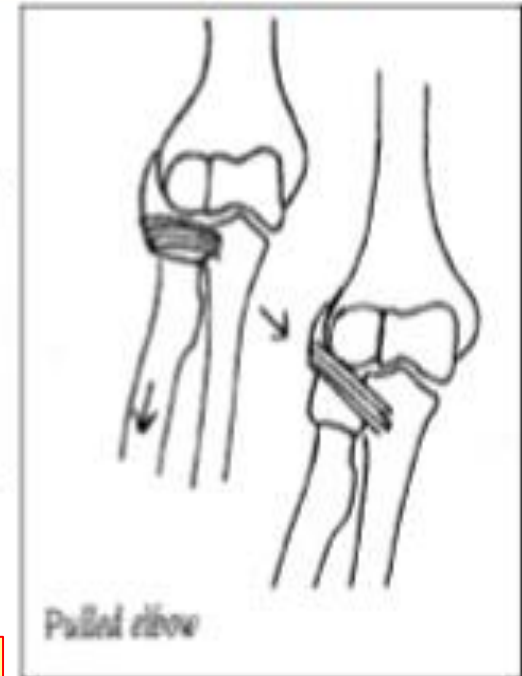
why preschool children (1-3y) are vulnerable to radial

head subluxation(pulled elbow)?

- Children are more prone to pulled elbow (nursemaid's elbow) because the shape of annular ligament is tubular and the head of radius is not fully developed(smaller)
- Therefore when he is suddenly lifted up & his arm pronated, the head of radius slips out partially from annular ligament result in subluxation of radius head
- Reduction done by **applying direct pressure posteriorly on head of radius simultaneously supinating & extending the forearm**



supination



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